1
Stellar Pressure and Density

1.1 *What is a Star?*

For our purposes, we start by defining a star as 'a self-gravitating mass of gas that radiates energy from an internal source'. The Sun, of course, is a star, and we use satellites such as Solar Dynamics Observatory to study its properties (see Figure 1.1). Stars are incredibly stable objects, and so in creating a model of a star we begin with hydrostatic equilibrium where thermal pressure inside the star opposes its gravitational collapse. As a consequence, this allows us to estimate the free-fall timescale for stellar collapse, and also the mean pressure of a star.

We then look at mass continuity, and estimate the mean density of a star. Finally we combine these equations to make an estimate of the minimum pressure in the core of a star. At the end of the lecture we apply these calculations to the Sun.

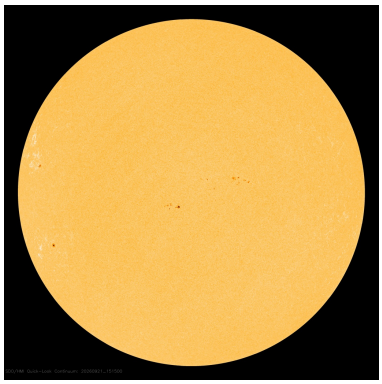

Figure 1.1: An image of the Sun taken by Solar Dynamics Observatory on 29 September 2026, and downloaded from https://sdo.gsfc.nasa.gov/data/

1.2 *Hydrostatic Equilibrium*

To explore pressure balance inside a star, we first make some assumptions, namely that the star is spherically symmetric (i.e. all variables only depend on r, the distance from the centre of the star), and also that the properties of the star vary slowly enough with time so that we can treat them as invariant. The star is defined to have radius R and total mass M.

Next, we define some unknown parameters. These are

- the total mass $m = m(r)$ **inside** radius r
- the density $\rho = \rho(r)$ **at** radius r
- the pressure $P = P(r)$ **at** radius r.

Note here that in each case we've explicitly stated the dependence of each parameter on r for clarity. It's also important to remember that $m(r)$ represents the mass **inside** the spherical volume of radius r!

We now use a 'pillbox analysis' to find the force balance inside the star, as illustrated in Figure 1.2. The pillbox has a thickness δr

and the area of both the top and bottom circular surface is A. It is placed at a distance $\boldsymbol{r} = r\hat{\boldsymbol{r}}$ from the centre of the star.

The density of the star depends on r only, because of spherical symmetry and so $\rho = \rho(r)$. The mass of the pillbox is therefore

$$\delta m = \rho(r) A \delta r \tag{1.1}$$

Two forces act on the pillbox: gravity and pressure.[1] The gravitational force, $\boldsymbol{F}_g$ acting on the pillbox is

$$\boldsymbol{F}_g = -\frac{Gm(r)\delta m}{r^2}\hat{\boldsymbol{r}}. \tag{1.2}$$

Substituting for equation 1.1 we find that

$$\boldsymbol{F}_g = -\frac{Gm(r)\rho(r)A\delta r}{r^2}\hat{\boldsymbol{r}}. \tag{1.3}$$

We now consider the pressure force. On the outer face, there is a force $P(r+\delta r)A$ acting inwards, in the $-\hat{\boldsymbol{r}}$ direction. On the inner face, there is a force $P(r)A$ acting outwards, in the $+\hat{\boldsymbol{r}}$ direction. Consequently, the total pressure force $\boldsymbol{F}_P$ is

$$\boldsymbol{F}_P = -P(r+\delta r)A\hat{\boldsymbol{r}} + P(r)A\hat{\boldsymbol{r}} \tag{1.4}$$

$$\boldsymbol{F}_P = -\left(\frac{P(r+\delta r)A - P(r)A}{\delta r}\right)\delta r\,\hat{\boldsymbol{r}} \tag{1.5}$$

$$\boldsymbol{F}_P = -\left(\frac{P(r+\delta r) - P(r)}{\delta r}\right)A\delta r\,\hat{\boldsymbol{r}} = -\frac{dP}{dr}A\delta r\,\hat{\boldsymbol{r}} \tag{1.6}$$

In equilibrium, the sum of the forces is zero, and there is no acceleration, i.e.

$$\Sigma\boldsymbol{F} = \boldsymbol{F}_g + \boldsymbol{F}_P = 0. \tag{1.7}$$

We can therefore write that

$$-\frac{Gm(r)\rho(r)A\delta r}{r^2} - \frac{dP}{dr}A\delta r = 0. \tag{1.8}$$

The δr terms cancel out, and rearranging we find that

$$\boxed{\frac{dP}{dr} = -\frac{Gm(r)\rho(r)}{r^2}.} \tag{1.9}$$

This is the first key equation making up our model of a star!

If we define g as the acceleration due to gravity, where

$$g = \frac{Gm(r)}{r^2}, \tag{1.10}$$

then if $r = R$, the star's radius, g is the star's surface gravity.[2] We can therefore write

$$\frac{dP}{dr} = -g\rho(r). \tag{1.11}$$

In this section, we have analysed the force balance inside the star based on assumptions of hydrostatic equilibrium. If pressure

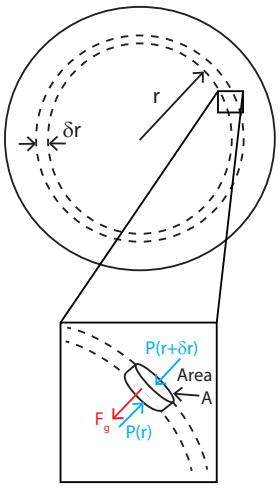

Figure 1.2: Diagram illustrating the pillbox analysis of hydrostatic equilibrium inside a star. If you are not familiar with this approach, please look back at the electrostatics part of Vector Fields Electricity and Magnetism in year 1.

[1] Two notes about the gravitational force: firstly looking at equation 1.2, we recall $m(r)$ is the mass inside r. Secondly, the gravitational force acts inwards, in the $-\hat{\boldsymbol{r}}$ direction.

[2] Use this equation to calculate the surface gravity of the Sun. The answer is given later in the notes.

is not equal to gravity, then there can be a net force, and therefore an acceleration, which could cause the star to expand or contract depending on the specific details.

1.2.1 *The Free-Fall Timescale*

Based on this analysis, we can start to compute some limiting properties. The first we will consider is the so-called free-fall timescale. Suppose the pressure inside the star disappears. Gravity remains, and so the star collapses. How long does this take? We can make an estimate as follows.

Consider the shell of material at the surface of the star. It must travel a distance R to the centre of the star. We estimate the acceleration as g, the gravitational acceleration at the surface. From basic mechanics, we know that under acceleration g over time t, an object starting at rest moves distance s where

$$s = \frac{1}{2} g t^2. \tag{1.12}$$

Therefore if $s = R$ and $g = GM/R^2$, and we write $t = \tau_{dyn}$ where τ_{dyn} is the free-fall time scale, we find that

$$R = \frac{1}{2} \frac{GM}{R^2} \tau_{dyn}^2 \tag{1.13}$$

Rearranging equation 1.13:

$$\tau_{dyn} = \left(\frac{2R^3}{GM} \right)^{1/2} \sim \left(\frac{R^3}{GM} \right)^{1/2}. \tag{1.14}$$

To simplify this expression and find the scaling behaviour, we have dropped the factor of $\sqrt{2}$ in the last step.[3]

[3] Use equation 1.14 to estimate the free-fall timescale of the Sun. The answer is given later in the notes.

1.2.2 *The Mean Pressure of a Star*

If pressure acts against gravity to hold a star in equilibrium, how big is that pressure, roughly speaking? To explore this, we can use the pressure balance equation, equation 1.9, to estimate the **mean** pressure inside a star. Clearly we do not expect the pressure to be uniform throughout the star, but this analysis shows how we can gain insight from even a very simple approach. We start by repeating the pressure balance equation:

$$\frac{dP}{dr} = -\frac{Gm(r)\rho(r)}{r^2} \tag{1.15}$$

We now need to estimate 'mean' values

- $M/2$ is our estimate[4] of the 'mean' $m(r)$
- $R/2$ is our estimate of the 'mean' r
- $\overline{P}$ is our estimate of the star's mean pressure

[4] Remember that this is the mass contained inside the radius. At the centre $m(0) = 0$ and at the edge $m(R) = M$, so our estimate is simply $M/2$.

Based on this, let's substitute $dP/dr \sim \overline{P}/R$, allowing us to write

$$\frac{\overline{P}}{R} \sim \frac{GM/2}{(R/2)^2}\overline{\rho} \qquad \Rightarrow \qquad \overline{P} \sim \frac{G}{2}\frac{M^2}{R^4}. \tag{1.16}$$

This equation provides a simple way to estimate the mean pressure inside a star.[5]

[5] Note that it is only an order-of-magnitude estimate, and so arguments about whether or not factors-of-two apply are not very important. What's more important is that this gives us some idea for how pressure scales with different parameters, and therefore the relative importance of different factors. In this way, we can use the equations to improve our understanding and intuition of the physics.

1.3 Mass Continuity

We have just found that the pressure gradient depends on m, ρ and r. However, they are not independent. This means that we need to introduce another equation. Consider Figure 1.3 which illustrates a spherical shell of mass δm, density $\rho(r)$, thickness δr and radius r inside the star. We can write that

$$\delta m = 4\pi r^2 \rho(r) \delta r. \tag{1.17}$$

Dividing by δr and taking the limit we find that

$$\boxed{\frac{dm}{dr} = 4\pi r^2 \rho(r).} \tag{1.18}$$

This is the second key equation of our model of a star![6] Integrating equation 1.18, we find that

$$m(r) = \int_0^r 4\pi\rho(r) r^2 dr. \tag{1.19}$$

Setting the upper limit of the integral as R gives, as expected, an expression for the total mass of the star, M_* where

$$M_* = \int_0^R 4\pi\rho(r) r^2 dr. \tag{1.20}$$

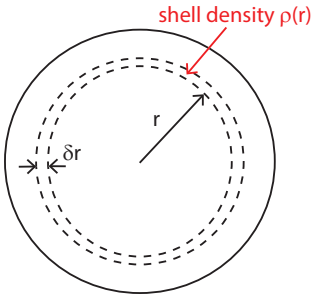

Figure 1.3: Illustration of a spherical shell inside a star.

[6] As ever, remember though that m is the mass **inside** r

1.3.1 The Mean Density of a Star

Clearly the density also varies inside the star, but it is helpful to estimate the mean density. If the mass of the star is M_* then the mean density[7] is $\overline{\rho}$ where

$$\overline{\rho} = \frac{M_*}{\frac{4}{3}\pi R^3}. \tag{1.21}$$

[7] Calculate the mean density of the Sun. The answer is given later in the notes.

1.4 Estimate of the Minimum Pressure at the Centre of a Star

In reality, the pressure varies through the star. It is of particular interest to estimate the pressure at the centre of the star, since this is where we expect it to be highest.[8] Here we make a minimum estimate (i.e. provide a lower limit) of the core pressure.

We now use both the pressure balance equation 1.9 and the mass continuity equation 1.18, which are repeated here:

$$\frac{dP}{dr} = -\frac{Gm(r)\rho(r)}{r^2} \tag{1.22}$$

[8] Remember though that it is the pressure gradient, not the absolute value of the pressure that balances the gravitational force.

$$\frac{dm}{dr} = 4\pi r^2 \rho(r) \tag{1.23}$$

Combining the two equations

$$\frac{dP/dr}{dm/dr} = -\frac{Gm(r)\rho(r)r^{-2}}{4\pi r^2 \rho(r)} \quad \Rightarrow \quad \frac{dP}{dm} = -\frac{Gm}{4\pi r^4} \tag{1.24}$$

Note that we have dropped the explicit dependence of m and P on r. We now integrate this equation, first writing

$$-dP = \frac{Gm}{4\pi r^4} dm. \tag{1.25}$$

The limits of the integral are from the core pressure P_c to the surface pressure P_s, and from $m(0) = 0$ to $m(R) = M_*$. Consequently

$$-\int_{P_c}^{P_s} dP = \int_0^{M_*} \frac{Gm}{4\pi r^4} dm \tag{1.26}$$

$$P_c - P_s = \int_0^{M_s} \frac{Gm}{4\pi r^4} dm \tag{1.27}$$

Since $r < R$, we know that $\frac{1}{r} > \frac{1}{R}$. We may therefore write that

$$\int_0^{M_*} \frac{Gm}{4\pi r^4}\, dm \; > \; \int_0^{M_*} \frac{Gm}{4\pi R^4}\, dm \; = \; \frac{GM^2}{8\pi R^4} \tag{1.28}$$

Finally, P_s is neglected[9], which allows us to write that

$$P_c \; > \; P_{\min} \; = \; \frac{GM^2}{8\pi R^4} \tag{1.29}$$

Previously in Section 1.2.2 we estimated the mean pressure of the Sun, whereas this is a more detailed estimate of the minimum pressure at the core of the Sun.

[9] We assume that the gas pressure at the surface is zero.

1.5 *Application to the Sun*

The properties of the Sun are often used as the reference values for the properties of other stars. The symbol $\odot$ is used as a subscript when referring to the Sun. Key numerical values include mass $M_\odot = 1.99 \times 10^{30}$ kg, and radius $R_\odot = 6.96 \times 10^8$ m. The gravitational constant is $G = 6.67 \times 10^{-11}$ Nm^{2}kg^{-2}.

Using equation 1.10, the surface gravity of the Sun is $g = 300$ ms^{-2}. The free-fall timescale of the Sun, using equation 1.14, is about 1,590 s. Equation 1.21 can be used to show that the solar mean density is approximately 10^3 kg m^{-3}. The mean density of the Sun is therefore similar to water.

Equation 1.29 shows that $P_{\odot,\min} = 4.5 \times 10^{13}$ N m^{-2}. The atmospheric pressure here on the surface of the Earth is about 10^5 N m^{-2}, and so our estimate is 10^8 greater than this. In fact our estimate is a strict lower limit, and the core pressure is even bigger than this.[10]

[10] Go back and make an estimate of the mean pressure using equation 1.16. You should find something rather surprising, which you should think about...

2
Stellar Plasma and Temperature

2.1 Stellar Plasma

In the previous lecture we established two equations, one for hydrostatic equilibrium and one for mass continuity, both of which we repeat here:

$$\frac{dP}{dr} = -\frac{Gm(r)\rho(r)}{r^2}, \tag{2.1}$$

$$\frac{dm}{dr} = 4\pi r^2 \rho(r). \tag{2.2}$$

Although there are 2 equations, we have 3 unknowns: the pressure $P(r)$, the mass inside radius r, $m(r)$, and the mass density $\rho(r)$. This means we need a further equation, an equation of state, to connect P and ρ. In many ways, the equation of state is where the physics lies because it is here that we need to start thinking about what stars are made of. In this lecture we first look at some properties of stellar plasma (plasma number density, mass fractions, and mean molecular weight), before considering stellar pressure and temperature.

In our model, we will treat the star as a fully-ionised plasma, in which the constituent atoms have been fully separated into ions and electrons. We will also treat the plasma as an ideal gas, although the true physics is evidently a lot more complicated.[1] However, we can get a surprisingly long way with these assumptions.

We start with a very basic question: what is the **number density** of the plasma inside a star? After deriving this, we will see how it is encoded in something called the mean molecular weight, μ, which provides information about the chemical composition of the star.

[1] Plasmas are wonderfully complex objects, and there is far more to say than could ever fit into this lecture course. If you are inspired to learn more, the year 3 plasma physics course and the year 4 space physics course are highly recommended!

2.1.1 Plasma Number Density

As before, we start with a series of assumptions:

- We assume that the star consists of three species, hydrogen, helium, and a 'metal', where the metal represents all the elements heavier than helium. The metal has electron number ℓ.

- We assume that the star is fully ionised. This means that

  $\mathrm{H} \rightarrow \mathrm{H}^{+} + \mathrm{e}^{-}$

  $\mathrm{He} \rightarrow \mathrm{He}^{2+} + 2\mathrm{e}^{-}$

  $\mathrm{metal} \rightarrow \mathrm{metal}^{\ell+} + \ell e^{-}$

Each hydrogen atom contributes 2 particles (a proton and an electron), each helium atom contributes 3 particles (an alpha particle and 2 electrons), and each metal atom contributes $\ell + 1$ particles (a nucleus and ℓ electrons). From this, we can add up the total number density n, i.e. the number of particles per unit volume, where

$$n = 2n_{\mathrm{H}^{+}} + 3n_{\mathrm{He}^{2+}} + (\ell + 1)n_{\mathrm{metal}^{\ell+}}. \tag{2.3}$$

2.1.2 *Mass Fractions*

In practice, we may know that the total mass of the star is M_{tot}, and that it consists of a fractional hydrogen mass X, fractional helium mass Y, and fractional metal mass Z. X, Y, and Z are referred to as mass fractions, where $X + Y + Z = 1$. We can write three equations:

$$X = \frac{M(\mathrm{H})}{M_{\mathrm{tot}}} \qquad X = \frac{M(\mathrm{He})}{M_{\mathrm{tot}}} \qquad Z = \frac{M(\mathrm{metal})}{M_{\mathrm{tot}}} \tag{2.4}$$

where $M(\mathrm{H})$, $M(\mathrm{He})$, and $M(\mathrm{metal})$ are the mass of hydrogen, helium and metal in the star, respectively.

First thinking about the hydrogen mass fraction, we can approximate the total hydrogen mass by neglecting the electrons and writing $M(\mathrm{H}) \sim N(\mathrm{H}^{+})m_H$. We find that

$$X \simeq \frac{N(\mathrm{H}^{+})m_H}{M_{\mathrm{tot}}} = \frac{N(\mathrm{H}^{+})m_H V}{M_{\mathrm{tot}} V} = \frac{n_{\mathrm{H}^{+}}}{\rho} m_H \tag{2.5}$$

In this step we multiplied by V/V where V is the volume of the star, allowing us to express X in terms of the number density and mass density. However, we want to express the number density in terms of the mass fraction. We therefore rearrange equation 2.5 to find

$$n_{\mathrm{H}^{+}} \simeq \frac{X\rho}{m_H}. \tag{2.6}$$

We now repeat this approach for the helium mass fraction, Y. In this case, $M(\mathrm{He}) \sim N(\mathrm{He}^{2+})m_{He}$. However, we also approximate $m_{He} \simeq 4m_H$, and so we can write

$$Y = \frac{M(\mathrm{He})}{M_{\mathrm{tot}}} \simeq \frac{N(\mathrm{He}^{2+})m_{\mathrm{He}}}{M_{\mathrm{tot}}} \simeq \frac{n_{\mathrm{He}^{2+}}}{\rho} 4m_H \tag{2.7}$$

Therefore

$$n_{\mathrm{He}^{2+}} \simeq \frac{Y\rho}{4m_H} \tag{2.8}$$

Finally, for the metal component, we assume that ℓ is large, and therefore that each metal atom has ℓ protons and ℓ neutrons. This means that $m_{metal} \simeq 2\ell m_H$. Following the same procedure,

$$Z = \frac{M(\text{metal})}{M_{\text{tot}}} \simeq \frac{N(\text{metal}^{\ell+}) m_{\text{metal}}}{M_{\text{tot}}} \simeq \frac{n_{\text{metal}^{\ell+}}}{\rho} 2\ell m_H. \tag{2.9}$$

Rearranging this equation,

$$n_{\text{metal}^{2\ell}} \simeq \frac{Z\rho}{2\ell_H}. \tag{2.10}$$

We now substitute equations 2.6, 2.8 and 2.10 into equation 2.3.

$$n \simeq 2\frac{X\rho}{m_H} + \frac{3}{4}\frac{Y\rho}{m_H} + \frac{(\ell+1)}{2\ell}\frac{Z\rho}{m_H} \tag{2.11}$$

$$n \simeq \left(2X + \frac{3}{4}Y + \frac{1}{2}Z\right)\frac{\rho}{m_H} \tag{2.12}$$

As a result, we are now able to express the number density in terms of the mass fractions X, Y, and Z.[2]

[2] Note that this is the mean number density in the star.

2.1.3 *Mean Molecular Weight*

We now use these ideas to define the mean molecular weight, μ, as follows:

$$\boxed{\mu = \frac{\rho}{m_H n}} \tag{2.13}$$

We now use equation 2.12 to substitute for n, and simplify the analysis by assuming that $Z = 0$. This means that $Y = 1 - X$, and therefore

$$\mu = \frac{\rho}{m_H n} = \frac{1}{2X + \frac{3}{4}Y} = \frac{4}{5X + 3}. \tag{2.14}$$

This equation allows us to calculate the mean molecular weight in terms of the mass fractions.

Up to now, we have presented the analysis in terms of the average properties over the whole star. However, the mass fractions could be different in different parts of the star. In this case, we can apply exactly the same analysis to part of the star, and compute local number densities and local mean molecular weights. For example, at the surface of the Sun X = 0.75 and Y = 0.25, so $\mu_{surface} \sim 0.6$. In the core of the Sun, $X \sim 0.4$, $Y \sim 0.6$, and $\mu_{core} \sim 0.8$.[3]

[3] Please see the problem sheet for the derivation of this equation when $Z \neq 0$.

2.2 *Stellar Pressure, Temperature and Equations of State*

In this section we now think about pressure and temperature. Inside a star, we must concern ourselves with both radiation pressure, from the light of the star itself, and thermal pressure, exerted by the stellar plasma. We start with some comments about thermodynamic equilibrium and equations of state.

2.2.1 *Thermodynamic Equilibrium and Equations of State*

In developing our understanding of how stars work, we will make assumptions relating to thermodynamic equilibrium because this allows us to introduce equations of state that relate pressure and temperature to density. Firstly, at a particular location in the star, it is reasonable to assume that collisions and mixing between the ions, electrons, and photons ensure all populations have the **same temperature**. Secondly, it is reasonable to assume that the star is in a statistically steady state, which means that bulk properties are constant (or at least changing very slowly with time). This means that time derivatives are zero (i.e. $\partial/\partial t = 0$).

A consequence of this is that quantities such as pressure and internal energy only depend on the mass density, temperature, and composition. However, the pressure, density and temperature vary through the star, so although we have an equilibrium at a particular location, it is local, and so we introduce the concept of local thermodynamic equilibrium.[4] With the assumption of thermodynamic equilibrium we can now introduce Equations of State to express the pressure in terms of temperature and the number density.[5]

[4] As a counter-example, the solar wind plasma is an example where there is not local thermodynamic equilibrium. In the solar wind, the temperature of the electrons and the ions are different.

2.2.2 *Radiation Pressure*

For a photon gas of temperature T, the radiation pressure is

$$P_{\text{rad}} = \frac{1}{3} a T^4 \tag{2.15}$$

where the radiation constant $a = 7.55 \times 10^{-16}$ J m^{-3} K^{-4}.

[5] The physics we use in this section will be discussed further in two different year 2 core modules, Quantum Physics and Thermal Physics & Structure of Matter. For the purpose of this module, we are not concerned with the derivation of these equations; we simply use them as part of the physics toolkit needed to understand stars without further analysis or derivation.

2.2.3 *Gas Pressure*

To calculate the gas pressure P_{gas} of the stellar plasma, we use the ideal gas law

$$P_{\text{gas}} = n k_B T \tag{2.16}$$

where n is the number density, k_B is the Boltzmann constant, and T is the plasma temperature. Rather than the number density n, we write this in terms of the mass density ρ, making use of the mean molecular weight μ and equation 2.14.

$$P_{gas} = n k_B T = \frac{\rho k_B T}{\mu m_H} = \frac{\rho \mathcal{R} T}{\mu} \tag{2.17}$$

In the last step we have substituted for the specific gas constant for hydrogen $\mathcal{R} = k_B/m_H$, where $\mathcal{R} \simeq 8300$ J kg^{-1} K^{-1}.

2.3 *Estimate of the Stellar Core Temperature*

To estimate the temperature in the core of a star, we start with the equation of state listed above, and set $P = P_c$, $T = T_c$, and $\rho \sim \overline{\rho}$ where $\overline{\rho}$ is the mean mass density. This allows us to write

$$P_c \sim \frac{\overline{\rho}\mathcal{R}T_c}{\mu} \tag{2.18}$$

We now use the hydrostatic equilibrium equation

$$\frac{dP}{dr} = -\frac{Gm(r)\rho(r)}{r^2} \tag{2.19}$$

and substituting for the nominal values given above, we find that

$$\frac{P_c}{R} \sim \frac{GM\overline{\rho}}{R^2}. \tag{2.20}$$

The next step is to eliminate P_c, by writing:

$$P_c \sim \frac{\mathcal{R}\overline{\rho}T_c}{\mu} \sim \frac{GM\overline{\rho}}{R} \quad \Rightarrow \quad T_c \sim \frac{\mu GM}{\mathcal{R}R} \tag{2.21}$$

An interesting consequence arises. If we assume that the mean density $\overline{\rho}$ is independent[6] of mass then $M \propto R^3$. This means that $R \propto M^{1/3}$, and $M/R \propto M^{2/3}$. Substituting for this expression we find that

[6] this is not completely true, $\overline{\rho}$ decreases with mass on the main sequence, i.e. more massive stars are less dense.

$$T_c \propto \frac{\mu GM^{2/3}}{\mathcal{R}} \tag{2.22}$$

Consequently, as M increases we expect that the core temp also increases.

For the Sun, we assume[7] that $\mu \sim 1$ and therefore[8]

$$T_{c,\odot} \sim \frac{GM_\odot}{\mathcal{R}R_\odot} \tag{2.23}$$

[7] This is to the nearest order of magnitude, a statement often heard in astrophysics!

[8] Evaluate T_c for solar parameters

In fact, more accurate solar models result in a temperature of 1.6×10^7 K. We may immediately deduce that some interesting physics must result, and we explore this next.

3
Stellar Energy and Fusion

3.1 Gravitational Energy

In this lecture we examine the question 'what powers stars?'. As a cloud of gas collapses to form a star, the conversion of gravitational potential energy causes the gas to heat up. This is described by the Virial theorem which we explore in some detail. However, we will find that the available energy is insufficient to explain what we observe. The answer is that the density and pressure in the core becomes high enough to allow nuclear fusion, which ultimately powers stars.

3.1.1 Gravitational Potential Energy of a Spherical Mass

We start by considering a spherical mass m, with radius r. We now add an infinitesimal mass dm which is brought from infinity to r. The work done is given by force $\times$ distance which then allows us to calculate $d\Omega$, the change in gravitational potential energy associated with the addition of mass dm.[1]

[1] For discussions about work and potential energy please look back at year 1 Mechanics.

$$d\Omega = \int_{\infty}^{r} \frac{Gm}{r^2}\, dm\, dr = -\frac{Gm\, dm}{r} \tag{3.1}$$

To calculate the gravitational potential energy of the whole object, which has mass M, we integrate over m.

$$\Omega = -\int_{0}^{M} \frac{Gm}{r} dm \tag{3.2}$$

3.1.2 The Virial Theorem

If we think about a cloud of gas collapsing to form a star, then gravitational potential energy is released. Where does it go? Thinking about the star, we start with hydrostatic equilibrium, and combine it with mass continuity to write

$$dP = -\frac{Gm}{4\pi r^4} dm \tag{3.3}$$

Now multiply both sides by V, volume, and integrate both P and m from the centre (subscript c) to the surface (subscript s) of the

star

$$\int_{P_c}^{P_s} V dP = -\int_0^{M_s} \frac{4\pi r^3}{3} \frac{Gm}{4\pi r^4} dm \tag{3.4}$$

We are going to show that the right hand side of this equation is related to the total gravitational potential energy of the system and the left hand side is related to the total internal energy.

First we consider the right hand side of equation 3.4.

$$-\int_0^{M_s} \frac{4\pi r^3}{3} \frac{Gm}{4\pi r^4} dm = -\frac{1}{3}\int_0^{M_s} \frac{Gm}{r} dm = \frac{1}{3} E_p \tag{3.5}$$

Here E_p is gravitational potential energy of the star.[2]

[2] Note we have changed terminology, E_p and Ω are the same.

Now we look at the left hand side of equation 3.4. Integrate by parts as both P and V are variables.

$$\int_{P_c}^{P_s} V dP = [PV]_c^s - \int_0^{V_s} P dV = (P_s V - P_c 0) - \langle P \rangle V \simeq -\langle P \rangle V \tag{3.6}$$

Here we have neglected P_s, because the we can reasonably assume that the surface pressure is a lot less than the average pressure.[3] Taking both results together, we find that

[3] Because V is treated as a variable, the value of V at the centre is 0.

$$\langle P \rangle V = -\frac{1}{3} E_p. \tag{3.7}$$

This is the Virial theorem which relates the average pressure $\langle P \rangle$ needed to support a self-gravitating system to its total gravitational energy E_p and volume V. The implication is that such a system will expand/contract until equation 3.7 is met. It is another way to think about the fact that a system will collapse or expand until pressure balances gravity, but here expressed in terms of energy.

Going one step further, the internal energy density of an ideal gas is $u = E_{th}/V = P/(\gamma - 1)$ where $\gamma = 5/3$.[4] Substituting for this expression we find that

[4] This will be discussed further in Thermal Physics and Structure of Matter.

$$\langle P \rangle V = \langle (\gamma - 1) u \rangle V = \frac{2}{3} \langle u \rangle V = \frac{2}{3} E_{th}. \tag{3.8}$$

Comparing this result with equation 3.7, we conclude that the negative gravitational potential energy is $2\times$ the thermal energy[5]

[5] At this point we have dropped writing the brackets to clarify the presentation.

$$-E_p = 2E_{th} \tag{3.9}$$

3.1.3 *Gravitational Contraction of a Star*

This results in an important insight. The total energy E_{tot} is the sum of the potential energy E_p and the thermal energy E_{th}.

$$E_{tot} = E_{th} + E_p = -\frac{1}{2} E_p + E_p = \frac{1}{2} E_p = -E_{th} \tag{3.10}$$

Thinking about how a star forms, in our model we begin with the material in a cloud distributed to infinity, and at the start $E_{tot} = E_p \simeq 0$. As the star collapses, $\Delta E_p < 0$, $\Delta E_{tot} < 0$, and $\Delta E_{th} > 0$. But, as the star forms it only needs half of the gravitational potential energy to be converted to thermal energy to stop

gravitational collapse! Where does the other half go? The Virial theorem says this must be radiated by the star. So we have a rather strange conclusion which is that as the star forms, it both radiates energy and the thermal energy goes up.

What happens if the cloud can't radiate energy? In this case Virial says that the cloud cannot collapse, because if it did, it would immediately heat up too much, and that pressure would counteract the self-gravitation of the cloud. There has to be a physical mechanism for energy to leave the system, otherwise the particles that make up the star can never slow down enough to form it.[6]

[6] It's important to emphasise here that the Virial theorem is not easy to understand! I remember I did not find it straightforward as a student, and it requires some thinking about.

3.1.4 *Estimating the Minimum Mean Temperature of a Star*

We now use the Virial theorem to estimate a minimum mean temperature of the Sun. to do this we start with the Virial theorem written in the following form[7]:

[7] We explicitly note that radius depends on mass.

$$\int_0^V P dV = \frac{1}{3}\int_0^M \frac{Gm}{r(m)} dm \tag{3.11}$$

Manipulating the left hand side, we use the ideal gas equation $P = \mathcal{R}\rho T/\mu$ as follows:

$$\int_0^V P dV = \int_0^V \frac{\mathcal{R}T}{\mu}\rho dV = \int_0^M \frac{\mathcal{R}T}{\mu} dm = \frac{\mathcal{R}\overline{T}}{\mu} M \tag{3.12}$$

Here $M\overline{T} = \int_0^M T dm$.

Now we consider the right hand side of equation 3.11. r varies from 0 to R, so $1/r \geq 1/R$. This means we can write

$$\frac{1}{3}\int_0^M \frac{Gm}{r(m)} dm \geq \frac{1}{3}\int_0^M \frac{Gm}{R} dm = \frac{GM^2}{6R} \tag{3.13}$$

Combining these results, we find that

$$\frac{\mathcal{R}\overline{T}}{\mu} M > \frac{GM^2}{6R} \tag{3.14}$$

$$\overline{T} \geq \frac{GM\mu}{6R\mathcal{R}} \tag{3.15}$$

Equation 3.15 is a lower limit, and remember it is an estimate for the minimum mean temperature of a star.[8]

[8] Calculate this for the Sun if $\mu = 0.5$

3.1.5 *Luminosity and Lifetime of a Gravitationally Powered Star*

We are now in a position to consider the lifetime of a star, if gravity is the only available energy source. Stars radiate energy and this is characterised by L, the luminosity, measured in watts.

We can assume that the lifetime $\tau = E_{avail}/L$, where E_{avail} is the available energy, and L is the luminosity, the rate at which energy is radiated by the star. The Virial theorem says that the thermal energy reservoir is approximately GM^2/R.[9] The Kelvin-Helmholtz timescale τ_{th} is time it would take for a star to passively cool given the above assumptions, where

[9] Here we are looking at the scaling based on the analysis in the previous section and ignoring factors of two.

$$\tau_{th} = \frac{GM^2}{RL} \tag{3.16}$$

for the Sun, τ_{th} is of the order of 10^7 years, i.e. 10 million years, which is inconsistent with e.g. geological evidence. In fact, of course, the Sun is powered by nuclear fusion.

3.2 Nuclear Fusion

3.2.1 Energy from Nuclear Fusion

Before looking in more detail at the nuclear reactions that power the Sun, we make some simple calculations to show that fusion is a sufficient power source. We will see that hydrogen burning converts four protons (hydrogen nuclei) into one alpha particle (helium-4 nucleus). Since $E = mc^2$, the amount of energy released is proportional to the difference in rest mass, $\Delta m_{rest} = 0.03 m_H$. The fraction of mass that is converted to energy is therefore $0.03 m_H / 4 m_H = 0.00775$, and the energy released is $\Delta E = 0.03 m_H c^2 \sim 27 MeV$.

Scaling up this calculation, if a star is initially completely made of hydrogen, then the total energy that can be released in this way is approximately[10] $0.00775 Mc^2$ where M is the mass of the star. If the star has luminosity L, the nuclear timescale tau_{nuc} is

[10] ignoring electrons

$$\tau_{nuc} = \frac{\Delta M c^2}{L} = \frac{0.00775 MC^2}{L} \tag{3.17}$$

For the Sun $\tau_{nuc} \sim 10^{11}$ years.

3.2.2 Nuclear Fusion Reactions in the Sun

In this subsection we briefly outline some key information about the nuclear fusion reactions that power the Sun. We start with the **proton-proton** chain or *pp* chain. This is illustrated in Figure 3.1. The *pp* chain has three branches.

- *pp* branch I (85% of energy generation in the Sun): two protons form a deuteron and release a positron and an electron neutrino. The deuteron and a proton form a helion (helium-3 ion). Two helion particles then interact to form an alpha particle, releasing two protons.
- *pp* branch II (15% of energy generation in the Sun): this branch occurs at higher temperatures. An alpha particle and a helion form a beryllium-7 ion. The addition of an electron converts this into a lithium-7 ion with the release of an electron neutrino. Finally, a proton interacts with the lithium-7 ion to create two alpha particles.
- *pp* branch III (<1 % of energy generation in the Sun): this also occurs at higher temperatures, but the beryllium-7 ion interacts

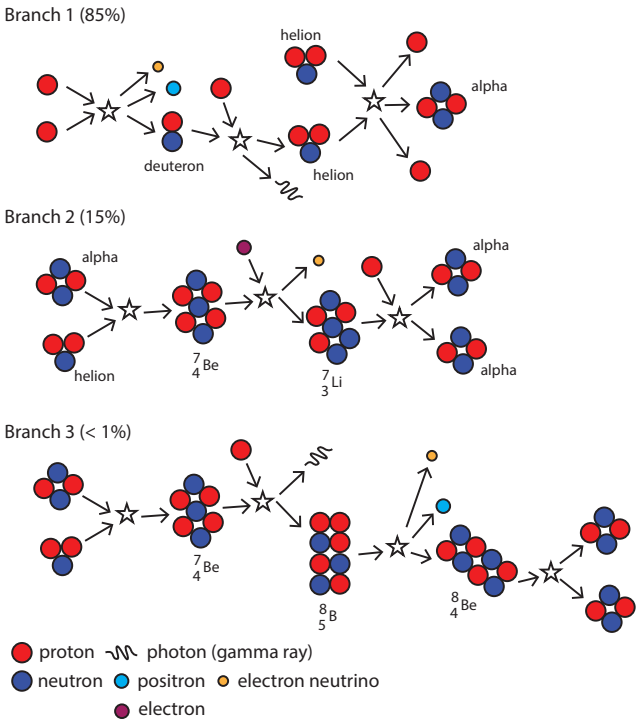

Figure 3.1: Cartoon summarising the three different branches forming the overall *pp* chain.

with a proton to form a boron-8 ion, emitting a gamma ray. The boron-8 ion emits an electron neutrino and a positron to form beryllium-8, which then breaks up into two alpha particles.

As the mass of a star increases, its core temperature increases, and the *pp* branch II/branch III reactions become more important.

The overall *pp* reaction converts four protons into an alpha particle, and this can be written down as[11]

$$\boxed{4\,{}^{1}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + 2e^{+} + 2\nu_e + 2\gamma} \tag{3.18}$$

where e^{+} is a positron, ν_e is an electron-neutrino, and γ is a gamma ray photon. This is an important equation which you should be able to recall.

[11] Strictly speaking, this is the branch I reaction, but this equation is typically used to summarise the *pp* reaction since branch I dominates.

At higher temperatures, a different reaction, the **carbon-nitrogen-oxygen** or *CNO* cycle becomes more important. This cycle is illustrated in Figure 3.2. Here carbon nuclei effectively act to catalyse the fusion of protons. A carbon-12 ion fuses with a proton to give a gamma ray and nitrogen-13, which then decays into carbon-13 by releasing a positron and electron neutrino. Carbon-13 fuses with a proton to give a gamma ray and nitrogen-14. Another proton fuses to give oxygen-15 and a gamma ray. The oxygen-15 then decays to nitrogen-15 releasing another positron/electron neutrino pair. Finally, nitrogen-15 fuses with a fourth proton to form an alpha particle and a carbon-12 ion which is then available to repeat the process.

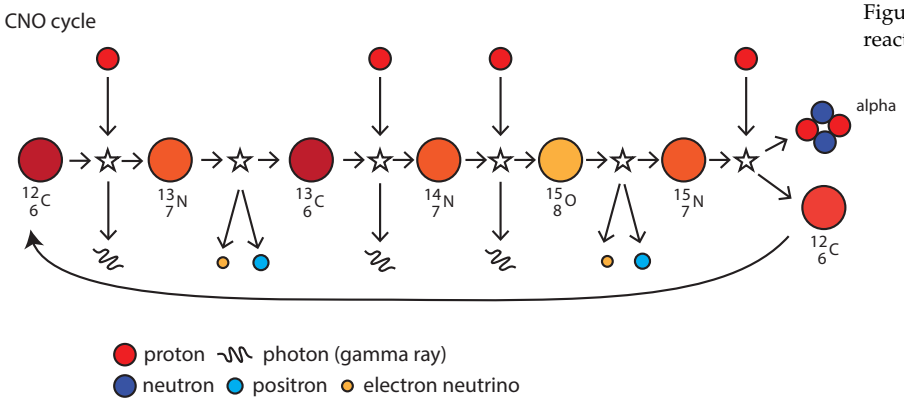

Figure 3.2: Illustration of the nuclear reactions forming the *CNO* chain.

3.2.3 *The Solar Neutrino Problem*

Neutrinos produced by nuclear fusion can be observed here on Earth. Their detection proved that fusion powers the Sun, and provides a window into the physics of the Sun and stars more generally.

However, a major surprise was the discovery that the flux of electron neutrinos is about one third to one half of that expected based on models of the solar interior. There are three flavours of neutrino (electron, muon and tau) and further experiments showed that the total number of solar neutrinos was consistent with internal models of the Sun, leading to the conclusion that neutrinos can change 'flavour', into muon- and tau-neutrinos.[12] This so-called neutrino oscillation implies that they have non-zero mass, which contradicts the current standard model. Consequently, new neutrino experiments such as DUNE, JUNO and Hyper-K are critical for pushing back our understanding of fundamental physics.[13]

[12] Solar neutrino research was awarded the 2002 and 2015 Nobel Prize in Physics.

[13] If you would like to know more about this, ask the Head of Dept., Prof. Stefan Söldner-Rembold!

3.2.4 *Rate of Energy Production*

The rate of energy production (in units of Wm^{-3}) depends on density and temperature. For the *pp* chain the energy production rate ϵ_{pp} is given by

$$\epsilon_{pp} \propto \rho T^4. \tag{3.19}$$

In contrast the energy production rate for the *CNO* cycle is ϵ_{CNO} where roughly speaking

$$\epsilon_{CNO} \propto \rho T^{20}. \tag{3.20}$$

Figure 3.3 illustrates how the energy production rate changes with temperature for each of the *pp* and *CNO* chains.

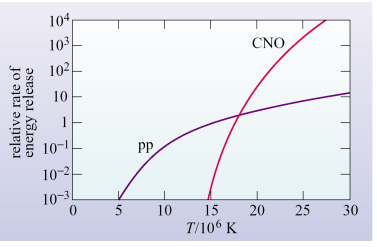

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Figure 3.3: The relative importance of the *pp* and the *CNO* chains in terms of their energy production rates. Figure 6.9 from An Introduction to the Sun and Stars, Simon F. Green & Mark H. Jones (eds), Cambridge University Press

4
Luminosity and the Transport of Energy

4.1 Luminosity and Energy Generation

In this lecture we examine luminosity in more detail and introduce the concept of opacity which is the resistance of a material to the passage of radiation. We then explore some different physical mechanisms by which energy is transported through stars.

As introduced in the previous lecture, luminosity L corresponds to the rate at which energy is radiated from an object. Although we often think about luminosity in terms of the properties of a star, it can be applied more generally. Here we develop an equivalent of the mass continuity equation, but for energy, to describe how energy is transported through a star.

As previously we assume spherical symmetry and radial transport. Consider a shell radius r as illustrated in Figure 4.1. Luminosity $L(r+\delta r)$ is defined as the total power radiated from the outer edge of the shell, and luminosity $L(r)$ is the total power entering the inner edge of the shell. Inside the shell energy is also being produced, and the rate of energy production/unit mass is ϵ. Consequently

$$L(r+\delta r) - L(r) = \epsilon \rho 4\pi r^2 \delta r, \tag{4.1}$$

$$\boxed{\frac{dL}{dr} = 4\pi r^2 \rho \epsilon.} \tag{4.2}$$

Equation 4.2 is the energy generation equation and is the **third key equation in our model of a star**.

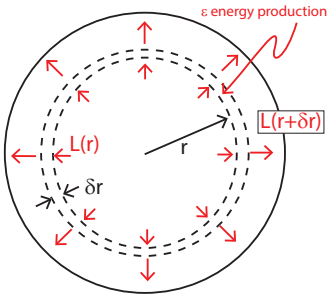

Figure 4.1: Cartoon showing the transmission and generation of energy through a thin shell inside a star.

4.2 Opacity and the Mean Free Path

Photons, i.e. radiation, do not generally pass freely through a material. Opacity quantifies this resistance of a material to the passage of radiation. As we will discover below, the physics of opacity can be very complicated, and depends on many material factors including density, temperature, and chemical abundance. However, we can simply define opacity as κ where

$$\kappa = 1/(\rho \ell). \tag{4.3}$$

Here ρ is the mass density and ℓ is the mean free path. The units of opacity are area/(unit mass). The mean free path is related to the concept of a random walk, illustrated in Figure 4.2. The photon continually interacts with matter, being scattered or absorbed/re-emitted in random directions. After N steps the photon has travelled a distance $\boldsymbol{r} = \boldsymbol{r}_1 + \boldsymbol{r}_2 + \cdots + \boldsymbol{r}_N$, and the root mean square displacement is

$$\langle \boldsymbol{r}^2 \rangle^{1/2} = \langle N \boldsymbol{r}_i^2 \rangle^{1/2} = \sqrt{N} \langle \boldsymbol{r}_i^2 \rangle^{1/2} = \sqrt{N} \ell \tag{4.4}$$

We now apply this to the Sun, where the mean free path $\ell \simeq 1$ mm. The distance from the centre to the Surface is $R_\odot$, and so using equation 4.4, this means that $N = (R_\odot/\ell)^2 = 10^{24}$ scattering events will occur. On average, the time taken for each step is $\Delta t = \ell/c$ where c is the speed of light. Consequently, the diffusion time, $\tau_{\rm diff}$, is

$$\tau_{\rm diff} = \frac{N\ell}{c} = \frac{R_\odot^2}{\ell c} \sim 10^5 \text{ years}. \tag{4.5}$$

The free-streaming time, which is the time it takes a photon to travel directly from the centre to the surface of the Sun, is $R_\odot/c \sim$ 2s. It therefore takes $10^{12}\times$ the free streaming time for a photon to exit the Sun.

There are four main processes that control opacity, illustrated in Figure 4.3. They are:

- Bound-bound absorption, which occurs when an atom absorbs an electron, moving the electron to a higher energy level. The atom can then radiate a photon in an arbitrary direction.
- Bound-free absorption. If the photon has sufficient energy, then it can be absorbed and cause a bound electron to escape its atom.
- Free-free absorption occurs when an electron (or indeed ion) absorbs a photon, increasing its velocity.
- Electron scattering corresponds to an elastic collision between a photon and an electron, causing the photon to change direction.

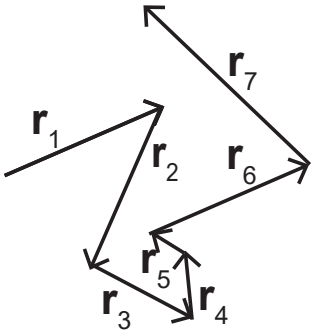

Figure 4.2: Illustration of a random walk

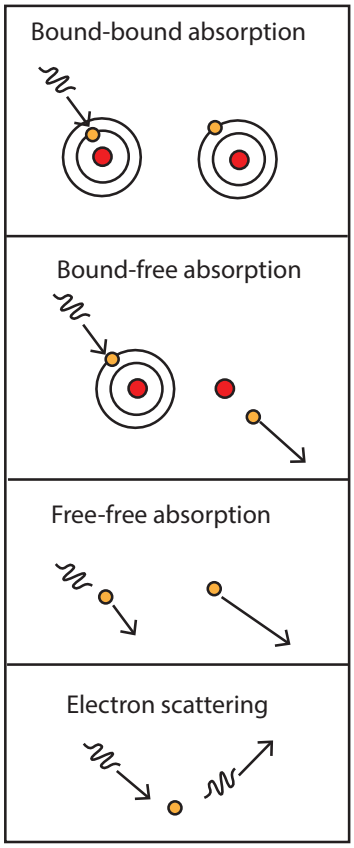

Figure 4.3: Four different physical processes that contribute to opacity.

4.3 *The Radiative Heat Transport Equation*

Here we derive an equation to describe the transport of heat by photons through a material. To create this equation, we will use Fick's law of diffusion. This states that if you have a gradient in particle density, ∇n, there will be a diffusive flux of particles j, connected by a coefficient of diffusion D where

$$j = -D\nabla n \tag{4.6}$$

and

$$D = \frac{1}{3} v \ell. \tag{4.7}$$

In the diffusion coefficient, v is the mean particle velocity and ℓ is the mean free path.[1]

To apply this concept to radiative diffusion, we think of energy density u and energy flux F instead of number density n and particle flux j. We replace n with u where u is the internal energy density of the photon gas. In parallel we replace the flux of particles j with the flux of energy F. We also replace v with c, the speed of light. This allows us to write

$$F = -\frac{1}{3}c\ell\nabla u. \tag{4.8}$$

Recall that previously we stated that $u = P/(\gamma - 1)$ (see the discussion before equation 3.8). We also found that $P_{rad} = (1/3)aT^4$ (equation 2.15). For a photon gas, $\gamma = 4/3$, and therefore

$$u = aT^4. \tag{4.9}$$

Taking the derivative of this equation (assuming spherical symmetry)

$$\frac{du}{dr} = 4aT^3\frac{dT}{dr}. \tag{4.10}$$

We then substitute this expression in equation 4.8, finding

$$F = -\frac{1}{3}c\ell 4aT^3\frac{dT}{dr}. \tag{4.11}$$

Finally, we use the expression for flux $F = L/(4\pi r^2)$, allowing us to write

$$\frac{L}{4\pi r^2} = -\frac{4ac\ell}{3}T^3\frac{dT}{dr}. \tag{4.12}$$

Substituting for $\kappa = 1/(\rho\ell)$, we arrive at the radiative transport equation

$$\boxed{\frac{dT}{dr} = -\frac{3\kappa\rho L}{16\pi acr^2T^3}} \tag{4.13}$$

This is one of two forms of our **fourth key equation describing a star**. It provides a relationship between the energy transport L, the opacity κ, and the temperature gradient $\frac{dT}{dr}$.

4.4 *Conductive Heat Transport*

For completeness, we briefly note that heat transport by particles can also occur. This is similarly described by the equations in the previous section, starting with Fick's law of diffusion. However, a different form of u will apply, as well as different values of v and ℓ. This type of transport is negligible in main sequence stars, but can be relevant in dense stellar remnants such as white dwarfs for example.

[1] These equations are a standard part of the physics toolkit which you may come across in other parts of the curriculum. Here we are using them as a standard result without further derivation or explanation.

4.5 Convective Heat Transport

Convective heat transport occurs through the bulk motion of macroscopic fluid elements. If the temperature gradient across the fluid is too large, the fluid becomes unstable, leading to the onset of convective transport through the bulk motion of macroscopic fluid elements.

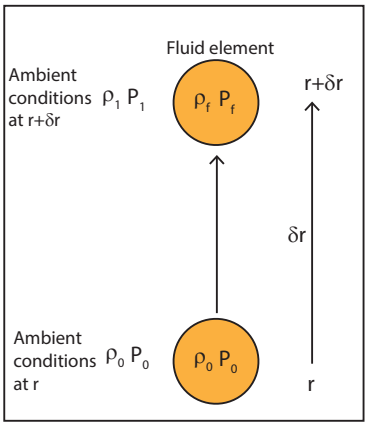

Figure 4.4: A cartoon illustrating the analysis of the convective instability.

4.5.1 The Schwarzschild Stability Criterion

Referring to Figure 4.4, at height r, $P(r) = P_0$ and $\rho(r) = \rho_0$. At height $r + \delta r$,

$$P_1 = P(r + \delta r) = P_0 + \delta r \frac{dP}{dr} \tag{4.14}$$

and

$$\rho_1 = \rho(r + \delta r) = \rho_0 + \delta r \frac{d\rho}{dr}. \tag{4.15}$$

We now displace the fluid element upwards. This displacement is **adiabatic**, meaning there is no heat transfer to or from fluid element. By definition[2]

$$\rho_f = \rho_0 \left(\frac{P_f}{P_0}\right)^{1/\gamma}. \tag{4.16}$$

[2] See the Thermal Physics and Structure of Matter Module.

However, $P_f = P_1$ because of pressure balance, so

$$\rho_f = \rho_0 \left(\frac{P_1}{P_0}\right)^{1/\gamma}. \tag{4.17}$$

Stability requires $\rho_f > \rho_1$ (i.e. element is more dense than its surroundings, and sinks, so

$$\rho_0 \left(\frac{P_1}{P_0}\right)^{1/\gamma} > \rho_0 + \delta r \frac{d\rho}{dr}. \tag{4.18}$$

Substituting for P_1, we find that

$$\rho_0 \left(1 + \frac{\delta r}{P_0}\frac{dP}{dr}\right)^{1/\gamma} > \rho_0 \left(1 + \frac{\delta r}{\rho_0}\frac{d\rho}{dr}\right) \tag{4.19}$$

Expanding this expression,

$$\rho_0 \left(1 + \frac{\delta r}{\gamma P_0}\frac{dP}{dr} + \dots\right) > \rho_0 \left(1 + \frac{\delta r}{\rho_0}\frac{d\rho}{dr}\right) \tag{4.20}$$

and so

$$\frac{\delta r}{\gamma P_0}\frac{dP}{dr} > \frac{\delta r}{\rho_0}\frac{d\rho}{dr} \tag{4.21}$$

In general

$$\frac{1}{\gamma P}\frac{dP}{dr} > \frac{1}{\rho}\frac{d\rho}{dr}. \tag{4.22}$$

This is the Schwartzschild stability criterion. If it is satisfied, i.e. if the left hand side is greater than the right hand side, then the system is stable.

We now use the equation of state $P = \mathcal{R}\rho T/\mu \propto \rho T$ to rewrite this equation in terms of T and P. The procedure is to first differentiate the equation of state as follows:

$$\ln P = \ln \rho + \ln T + const \tag{4.23}$$

$$\ln \rho = \ln P - \ln T + const \tag{4.24}$$

$$\frac{1}{\rho}\frac{d\rho}{dr} = \frac{1}{P}\frac{dP}{dr} - \frac{1}{T}\frac{dT}{dr}. \tag{4.25}$$

We then substitute this expression into the stability criterion equation 4.22, finding that

$$\frac{1}{\gamma P}\frac{dP}{dr} > \frac{1}{P}\frac{dP}{dr} - \frac{1}{T}\frac{dT}{dr}. \tag{4.26}$$

Rearranging this equation,

$$\left(\frac{1}{\gamma} - 1\right)\frac{dP}{dr} > -\frac{P}{T}\frac{dT}{dr}. \tag{4.27}$$

Since all the gradients are negative, we use absolute values to show that

$$\left(\frac{1}{\gamma} - 1\right)\left|\frac{dP}{dr}\right| > \left|\frac{P}{T}\frac{dT}{dr}\right| \tag{4.28}$$

The left hand side of equation 4.28 represents the adiabatic temperature gradient, and the right hand side represents the actual temp grad. If the actual temperature gradient is smaller, then the fluid is stable. If the actual temperature gradient exceeds the adiabatic gradient, it is unstable.

4.5.2 Heat Transport by Convection

In considering the transport of heat by convection we assume that macroscopic fluid elements rise a distance ℓ_{mix}, known as the mixing length. This distance is similar to the pressure scale height. Once the fluid element arrives, excess heat is then exchanged with its surroundings. Convective heat transport is very efficient, which means that we may assume that the system is at the limit of instability. That is to say, the actual temperature gradient grows until it reaches the adiabatic limit, but it will not grow larger because convection will start, redistribute the heat, reduce the heat gradient, and bring the system back to (marginal) stability. In conclusion

$$\boxed{\frac{dT}{dr} = \left(1 - \frac{1}{\gamma}\right)\frac{T}{P}\frac{dP}{dr}.} \tag{4.29}$$

This is the convective transport equation, the second of two forms of our **fourth key equation describing a star**.

5
The Stellar Structure Equations

5.1 Summary of the Equations of Stellar Structure

There are seven variables: pressure, P, mass, m, temperature, T, luminosity, L, density ρ, opacity, κ and energy generation rate, ϵ. These variables are connected by four differential equations and three other equations to provide closure.

The **differential equations** are:

1. Hydrostatic equilibrium (Lecture 1)

$$\frac{dP}{dr} = -\frac{Gm\rho}{r^2} \tag{5.1}$$

2. Mass continuity (Lecture 1)

$$\frac{dm}{dr} = 4\pi r^2 \rho \tag{5.2}$$

3. Heat transport: two options

   Radiation (Lecture 4)

$$\frac{dT}{dr} = -\frac{3\kappa\rho L}{16\pi a c r^2 T^3} \tag{5.3}$$

   Convection (Lecture 4/5)

$$\frac{dT}{dr} = \left(1 - \frac{1}{\gamma}\right)\frac{T}{P}\frac{dP}{dr} \tag{5.4}$$

4. Energy generation (Lecture 4)

$$\frac{dL}{dr} = 4\pi r^2 \rho \epsilon \tag{5.5}$$

The **closure equations** are:

1. Equation of state: two options

   Radiation pressure (Lecture 2)

$$P_{\rm rad} = \frac{1}{3} a T^4 \tag{5.6}$$

   Gas pressure (Lecture 2)

$$P_{\rm gas} = \frac{\mathcal{R}\rho T}{\mu} \tag{5.7}$$

2. Energy generation (Lecture 3)

$$\epsilon \propto \rho^{\alpha} T^{\eta} \tag{5.8}$$

3. Opacity (Lecture 4)

$$\kappa = \kappa_0 \rho^{\alpha} T^{\beta} \tag{5.9}$$

In the energy generation equation and the opacity equation, for our purposes $\alpha = 1$ and is constant. Regarding energy generation, as we saw in lecture 3, η=4 for hydrogen burning via the pp chain, and $\eta \simeq 16 - 20$ for the CNO cycle. Finally, in the opacity equation, $\beta = -3.5$ for low mass stars[1], whereas for high-mass stars opacity is constant.[2]

[1] The low-mass relation is called Kramer's opacity, and is discussed later in this lecture.

[2] In reality, α, η, and β contain a large amount of physics, and they are not constant, but depend on density, temperature and composition, among other things.

5.2 *Boundary Conditions*

Boundary conditions are needed to solve these equations and they are listed in Table 5.1. Note that in setting the outer boundary conditions for T, P, and ρ as zero, we are simply arguing that they are negligible compared to the higher values found inside the star.

Table 5.1: Boundary conditions at the centre and the surface of a star. Note that the surface temperature, pressure and density are considered negligible and set equal to 0.

Stellar Boundary Conditions		
Variable	$r = 0$	$r = R$
m	0	M_*
L	0	L_*
T	T_C	0
P	P_C	0
ρ	ρ_C	0

5.3 *The Structure of the Sun*

The standard solar model has been developed over many decades, by a number of different scientists. Of particular note is the work of John Bahcall.[3] Here we will look at a particular model published in 2001, BP2000, the data for which are available here.

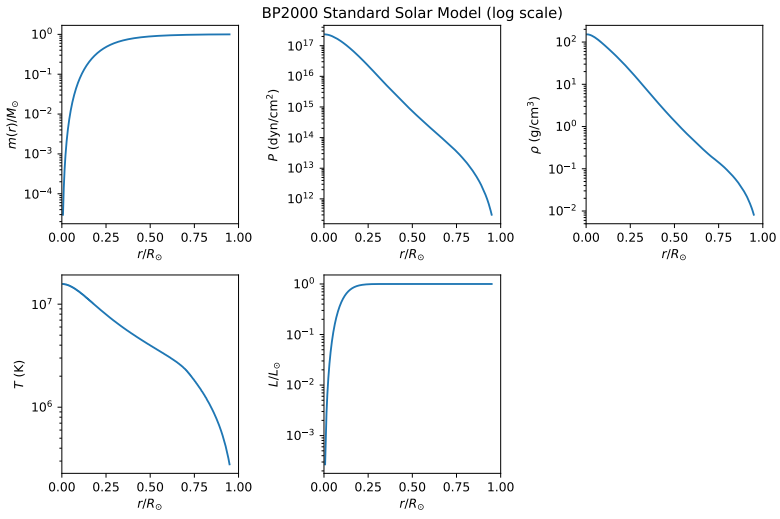

Figure 5.1: Plot showing the variation of enclosed mass m, Pressure P, density, ρ, temperature T, and luminosity L, based on the BP2000 model. A logarithmic scale is used on the y-axis. Underlying data are available here.

Figure 5.1 shows how the enclosed mass m rapidly increases close to the centre of the Sun, and then increases slowly through the outer layers. Notice that the luminosity L also rapidly increases, and is essentially constant above 0.3 $R_\odot$. The temperature, pressure, and density all decrease. Note how P and ρ both quickly decrease close to the surface. The temperature is slightly different, with a distinct change in the temperature gradient at around 0.7 $R_\odot$.

[3] John Bahcall worked at the Institute for Advanced Study. He passed away in 2005, but his website is still maintained here.

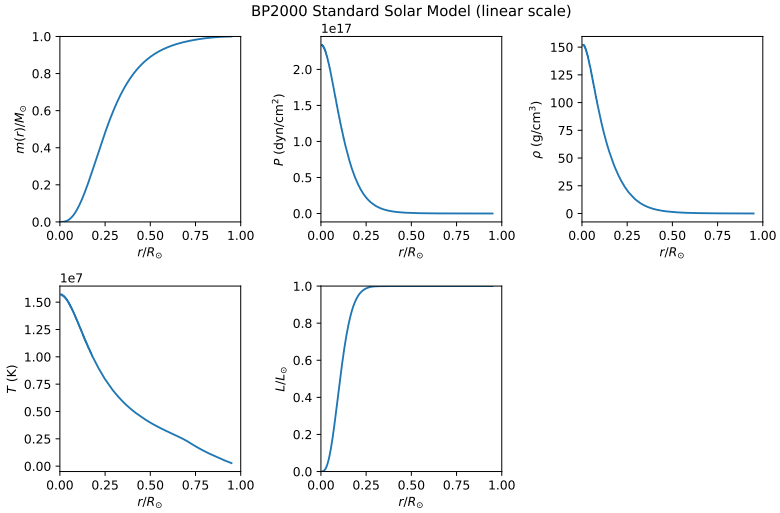

Figure 5.2: Plot showing the variation of enclosed mass m, Pressure P, density, ρ, temperature T, and luminosity L, based on the BP2000 model. This figure is in the same format as Figure 5.1 but a linear scale is used on the y-axis to highlight the core more clearly. Underlying data are available here.

Figure 5.2 shows the same data as Figure 5.1, but uses a linear scale on the y-axis. This shows us more clearly that below about 0.3 $R_\odot$, the density, pressure, and temperature are all substantially higher. This region is the core, where the vast majority of the Sun's luminosity is generated by nuclear fusion.

The physics of the core is further revealed by Figure 5.3, which shows the mass fraction of hydrogen, X, and helium, Y, as a function of radius. On average, the Sun is 73% hydrogen, 25 % helium and 2% metal.[4] Below 0.3 $R_\odot$, the fraction of hydrogen is reduced, and the fraction of helium is increased. The depletion of hydrogen is caused by billions of years of nuclear fusion.

Outside the core, the luminosity is essentially constant, and so here energy is being transported through the layers to the surface. However, Figure 5.3 shows that there is a change in the mass fractions at $r = 0.714 R_\odot$, marked by the vertical dashed line. This is because of a change in the physics by which the energy is being transported. Below this radius, radiative transport dominates. Above this radius, convection dominates. The Sun is therefore conventionally divided into three regions - the core (where energy is generated), the radiative zone (where radiative transport dominates), and the convection zone (where convection dominates). The onset of convection is caused by the fact that eventually the temperature gradient exceeds the Schwartzschild criterion. Referring back to Figure 5.1, we can now see that convection actually allows the temperature gradient to become a lot steeper in the convective zone.

[4] If you follow the link to the underlying data, you can also plot the mass fraction of some other constituents.

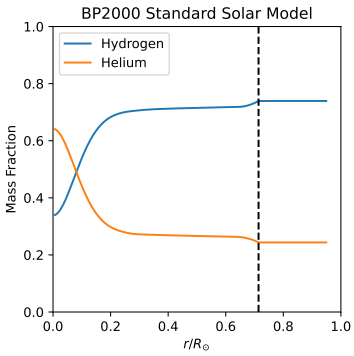

Figure 5.3: Plot showing the variation of hydrogen mass fraction (X) and helium mass fraction (Y) as a function of radius, based on the BP2000 model. Underlying data is available here.

5.4 *Further Comments on the Sun and its Stability*

The three-part structure of the Sun is illustrated in Figure 5.4. Nuclear fusion in the core is the source of the energy that the Sun ultimately radiates; this energy is transported to the surface first by radiative transport and then by convection. A particularly powerful

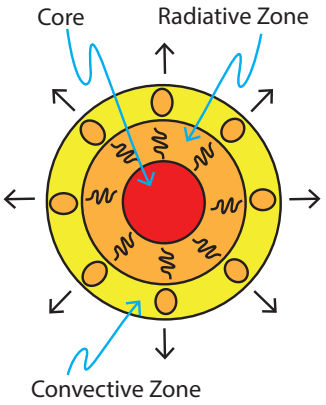

Figure 5.4: Cartoon illustrating the three different parts of the solar interior.

technique for probing the interior of the Sun is helioseismology. The Sun oscillates with a variety of acoustic modes, which penetrate to different depths depending on their wavelength. By measuring the Doppler shifts associated with this motion, it is possible to infer the internal structure in considerable detail.[5]

[5] See for example, research at the University of Birmingham.

The Sun has existed for billions of years because its dynamics are fundamentally stable as we now describe. Suppose that energy production increases in the core. This would increase the Sun's potential energy according to the Virial theorem, making it less negative. As a consequence the Sun would expand, in turn causing it to cool. This cooling would then reduce the energy production rate. On the other hand, if the energy production decreases then the Sun would contract, which would increase its temperature and its energy production rate.[6]

[6] In this way, the Sun is a stable, gravitationally-confined, fusion reactor. In contrast, plasma confinement by either inertial or magnetic forces is extremely unstable, which makes fusion power on Earth challenging.

5.5 *Stars in General: Homology Transformations*

Now that we have considered the Sun, what about other stars? In principle, we could of course simply attempt to solve the equations for each star we observe. But, we can also consider more elegant approaches. For example, in star clusters (see Figure 5.5), we know that all the stars are made of roughly the same material, and so will have the same composition. They are a homologous group. Homology arguments provides a way to immediately estimate the properties of all the stars, by constructing a single dimensionless version of the stellar structure equations (the homology solution), and scaling the solution to each star in the based on its mass.

Figure 5.5: The NGC 6440 globular cluster, observed by the James Webb Space Telescope Near-InfraRed Camera (NIRCam). Credit: ESA/Webb, NASA & CSA, P. Freire, Acknowledgement: M. Cadelano and C. Pallanca.

To do this, each variable must be dimensionless. Firstly, instead of r we use $x = r/R$ where R is the radius of the star. Using pressure as an example, we then write $P = P_c P_x(x)$ where $P_x(x)$ is the dimensionless functional form, and P_c is a scaling factor (here, it would be the pressure in the core). $P_x(x)$ is the same for all stars, and P_c is unique to each star. We repeat this for all the variables, and this allows the equations to be re-written in dimensionless form. We then solve those equations, and apply the appropriate scaling factors to find the bespoke solution for each star.

Homology is not without its limitations because it makes one very large assumption, namely that the physics is independent of mass. So whilst useful for exploring the likely behaviour of different star populations, it must be used with care, because it may lead you astray. For example, it can be shown that for stars similar to the Sun, the outer convection zone is bigger for lower-mass stars and smaller for higher-mass stars. If the mass is sufficiently high, the core temperature becomes high enough that the CNO cycle dominates the energy release, and this causes a much higher temperature gradient in the core. This is unstable to the onset of convection, and so high-mass stars have a convective core surrounded by a radiative envelope. This physical insight would not be captured by homology arguments.

6
Stellar Scaling Relations

6.1 Core Density and Pressure

In this lecture we consider the scaling relations between different stellar parameters, starting with the relationship between core density and pressure. We expect that the core density ρ_c is proportional to the total mass M divided by R^3:

$$\boxed{\rho_c \propto \frac{M}{R^3}} \tag{6.1}$$

We now use the hydrostatic equilibrium equation,

$$\frac{dP(r)}{dr} = -\frac{Gm\rho}{r^2}. \tag{6.2}$$

The gradient $dP/dr \sim (P_s - P_c)/(R - 0) = -P_c/R$. Substituting for M and R, and dropping constants, we can write that

$$\frac{P_c}{R} \propto \frac{M\rho_c}{R^2} \propto \frac{M^2}{R^5} \tag{6.3}$$

$$\boxed{P_c \propto \frac{M^2}{R^4}} \tag{6.4}$$

Equation 6.4 is the scaling relation describing how the core pressure depends on the mass and radius of the star.

6.2 Core Temperature

The scaling relation for the core temperature starts with the equation of state

$$P = \frac{\mathcal{R}\rho T}{\mu} \tag{6.5}$$

Rearranging this equation, substituting for T_c, P_c and ρ_c and dropping the constants, we find

$$T_c \propto \frac{P_c}{\rho_c} \tag{6.6}$$

We then substitute in equations 6.1 and 6.4:

$$T_c \propto \frac{M^2}{R^4}\frac{R^3}{M} \tag{6.7}$$

$$\boxed{T_c \propto \frac{M}{R}} \tag{6.8}$$

This shows that the core temperature T_c simply scales with the ratio of M/R.

6.3 *Scaling of Luminosity with Mass*

To investigate how luminosity scales with mass, we start with the radiative heat transport equation, as this contains L:

$$\frac{dT}{dr} = -\frac{3\kappa\rho L}{16\pi acr^2T^3}. \tag{6.9}$$

We now substitute in the expression for opacity

$$\kappa(\rho, T, X) = \kappa_0\rho^\alpha T^\beta, \tag{6.10}$$

finding that

$$\frac{dT}{dr} = -\frac{3\kappa_0\rho^\alpha T^\beta\rho L}{16\pi acr^2T^3}. \tag{6.11}$$

We now substitute for T_c, R, L_*, and ρ_c, and drop constants as before:

$$\frac{T_c}{R} \propto \frac{\rho_c^\alpha T_c^\beta \rho_c L_*}{R^2T_c^3} \tag{6.12}$$

Rearranging this equation to find an expression for L_*, we find that

$$L_* \propto R^2T_c^3\frac{T_c}{R}\frac{1}{\rho_c^{1+\alpha}T_c^\beta} \tag{6.13}$$

$$L_* \propto \frac{T_c^{4-\beta}R}{\rho_c^{1+\alpha}} \tag{6.14}$$

Now we substitute for T_c, using equation 6.8, and for ρ_c, using equation 6.1, to find

$$L_* \propto R\left(\frac{M}{R}\right)^{4-\beta}\left(\frac{M}{R^3}\right)^{-1-\alpha}, \tag{6.15}$$

The exponent of M is $4-\beta-1-\alpha = 3-\beta-\alpha$, and the exponent of R is $1-4+\beta+3+3\alpha = 3\alpha+\beta$. We therefore find that

$$L_* \propto M^{3-\beta-\alpha}R^{3\alpha+\beta}. \tag{6.16}$$

For high mass stars, it is reasonable to assume that the opacity κ is constant, and therefore independent of temperature and density. In this case, $\alpha = \beta = 0$, and we find that

$$\boxed{L_* \propto M^3} \tag{6.17}$$

The prediction that L_* scales with the cube of the mass M for high mass stars is a important result. Figure 6.1 plots the relationship between L_* and M for a population of stars, and in fact for

higher mass stars above 5 $R_\odot$ equation 6.17 is a reasonable fit. Note that the x-axis is reversed so that mass increases to the left! More generally, a reasonable choice based on observations is

$$\boxed{L_{*,obs} \propto M^{3.5}} \tag{6.18}$$

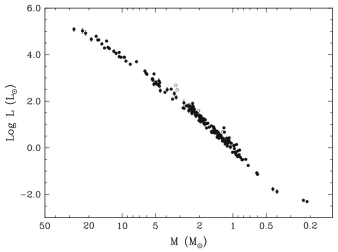

Figure 6.1: A plot of luminosity as a function of mass for the population of stars surveyed by Torres et al., Astron. Astrophys. Rev., 18, 67-126, 2010. This is a reproduction of Figure 5. For higher mass stars above 5 $M_\odot$, the luminosity is proportional to M^3.

6.4 Radius - Mass Relation

Two of the stellar structure equations contain L, and so this provides a way to find how the radius R scales with mass M. We start with the energy transport equation

$$\frac{dL}{dr} = 4\pi r^2 \rho\epsilon \tag{6.19}$$

The energy production rate η is given by

$$\epsilon = \epsilon_0 \rho T^\eta, \tag{6.20}$$

where η depends on the underlying nuclear reactions (recall η = 4 for the pp chain and we use $\eta \sim 16 - 20$ for the CNO cycle). Substituting for L_*, R, ρ_c and T_c,

$$\frac{L_*}{R} \propto R^2 \rho_c^2 T_c^\eta. \tag{6.21}$$

We now substitute for T_c using equation 6.8 and ρ_c using equation 6.1.

$$\frac{L_*}{R} \propto R^2 \frac{M^2}{R^6} \frac{M^\eta}{R^\eta}, \tag{6.22}$$

$$L_* \propto \frac{M^{2+\eta}}{R^{3+\eta}}. \tag{6.23}$$

But, if $L_* \propto M^3$ then

$$M^3 \propto \frac{M^{2+\eta}}{R^{3+\eta}} \tag{6.24}$$

$$R^{3+\eta} \propto M^{\eta-1} \tag{6.25}$$

If η = 4 (low mass stars), then

$$R^7 \propto M^3 \qquad \rightarrow \qquad R \propto M^{3/7}. \tag{6.26}$$

If η = 16 (high mass star), then

$$R^{19} \propto M^{15} \qquad \rightarrow \qquad R \propto M^{15/19} \sim M^{3/4} \tag{6.27}$$

Comparing with Figure 6.2, the scaling relationship for high-mass stars given by equation 6.27 is in surprisingly good agreement with the observations. At intermediate-masses the observed scaling is slightly flatter, before becoming steeper for low-mass stars. In the low-mass population a scaling close to R ∝ M is in fact seen.

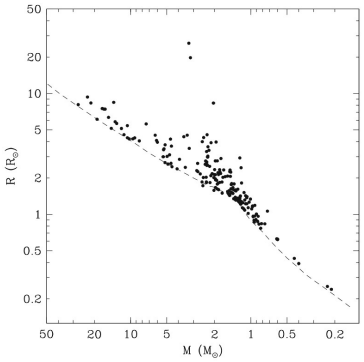

Figure 6.2: A plot of radius as a function of mass for the population of stars surveyed by Torres et al., Astron. Astrophys. Rev., 18, 67-126, 2010. This is a reproduction of Figure 2. For higher mass stars, the radius is proportional to $M^{3/4}$.

6.5 Stellar Surface Temperature

Up to now, we have approximated the surface temperature of stars as 0, as it is much lower than the temperature in the core. However, this is evidently not true, and so here we explore how to quantify the surface temperature. To do this we use the Stefan-Boltzmann equation[1], which equates the energy flux (units Wm^2) from a black-body, F, to its effective temperature, T_{eff}, where

[1] See Thermal Physics and Structure of Matter and Quantum Physics

$$F = \sigma T_{eff}^4 \tag{6.28}$$

We can write the flux F in terms of the luminosity L_*,

$$F = \frac{L_*}{4\pi R^2}. \tag{6.29}$$

Consequently,

$$L_* = 4\pi R^2 \sigma T_{eff}^4 \tag{6.30}$$

For **low-mass stars**,

$$R \propto M^{3/7} \propto \left(L_*^{1/3}\right)^{3/7} = L_*^{1/7}. \tag{6.31}$$

Rearranging equation 6.30, and substituting for R, we find

$$T_{\text{eff}}^4 \propto \frac{L_*}{R^2} \propto \frac{L_*}{L_*^{2/7}} \propto L_*^{5/7} \tag{6.32}$$

$$L_* \propto T_{eff}^{28/5}. \tag{6.33}$$

For **high mass stars**, $R \propto M^{3/4} \propto L_*^{1/4}$, and

$$T_{\text{eff}}^4 \propto \frac{L_*}{L_*^{1/2}} \tag{6.34}$$

$$L_* \propto T_{\text{eff}}^8. \tag{6.35}$$

This means that we expect to see the scaling between luminosity and temperature change for higher mass stars where the CNO cycle becomes important.[2]

[2] We will return to this point later in the module, when we introduce and discuss the Hertzsprung-Russell diagram.

6.5.1 Stellar Lifetimes

The lifetime depends on the available energy which is proportional to M. If L_* is the rate of energy loss, and $L_* \propto M^{-3.5}$, then the lifetime of such stars is

$$\tau \propto \frac{energy}{L_*} \propto \frac{M}{L_*} \propto M^{-2.5}. \tag{6.36}$$

This is a rather counter intuitive result, because more massive stars have a shorter lifetime.[3]

[3] If the lifetime of the Sun is expected to be 10 bn years, estimate the lifetime of a 0.5 $M_\odot$ and a 5 $M_\odot$ star.

7
Observing the Sun, Our Nearest Star

7.1 *The Solar Flux Spectrum*

Figure 7.1 shows the irradiance, or flux, from the Sun as a function of wavelength. We observe that it peaks at around a wavelength of 500nm, and that it falls off smoothly as wavelength increases. At shorter wavelengths the irradiance is also reduced, and it is much more structured. The overall shape of the spectrum can be explained in terms of blackbody radiation together with absorption and emission associated with atomic spectral lines. This in turn allows us to explore the detailed structure of the solar atmosphere.

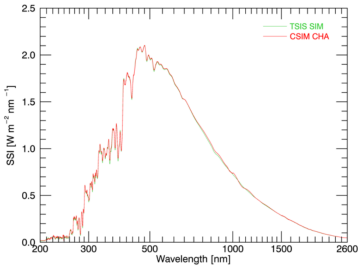

Figure 7.1: Measurements of the Solar Spectral Irradiance, reported by Richard et al., 2019, Figure 19. CSIM is the Compact Spectral Irradiance Monitor, a 6U CubeSat mission, and TSIS is an instrument hosted on the International Space Station.

7.2 *Blackbody (Thermal) Radiation*

As discussed in previous lectures, at the surface of the Sun, the density drops further, the opacity reduces, and the light from the Sun escapes. The layer of the Sun where the photons decouple from the plasma is called the photosphere.[1]

Observations of Sun's intensity as a function of wavelength prompted one of the most important revolutions in physics. A perfect 'blackbody' absorbs all radiation falling on it, which means that it heats up. It therefore also radiates, to remain in thermal equilibrium. The radiation from a blackbody is 'thermal', and in fact stars are excellent examples of blackbodies, because the photon and particle populations are tightly coupled by collisions.[2] Classical physics predicts that this radiation flux is infinitely large at short wavelengths[3]; Planck proposed that energy was absorbed or emitted in discrete packets, or quanta. This directly leads to Planck's function which correctly describes the intensity distribution B_ν of radiation from a blackbody:

$$B_\nu(T)d\nu = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/k_BT}-1}d\nu, \tag{7.1}$$

where h is Planck's constant, k_B is Boltzmann's constant, c is the speed of light, ν is frequency, and T is the temperature of the blackbody. The units of B_ν are power/ unit area/ unit solid angle/ unit frequency. When making measurements, we in fact measure B between two frequencies, and so we integrate $B_\nu(T)d\nu$ from ν_1 to

[1] Most light comes from the lower photosphere.

[2] Recall there are many collisions and interactions between the photons and the plasma particles in the Sun, ensuring that Local Thermodynamic Equilibrium (LTE) is maintained. Therefore the temperature of the photon gas is the same as the plasma.

[3] This is the so-called ultraviolet catastrophe.

ν_2. To convert to wavelength λ, we take the derivative of the wave equation, $c = \nu\lambda$, writing

$$d\nu = \frac{c}{\lambda^2} d\lambda. \tag{7.2}$$

Here we may ignore the minus sign, as it is accounted for in the limits of the integral, that is if $\nu_2 > \nu_1$ then $\lambda_2 < \lambda_1$. Substituting for $d\nu$ in equation 7.1,

$$B_\nu(T)d\nu = B_\nu(T)\frac{c}{\lambda^2}d\lambda = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/k_BT} - 1}\frac{c}{\lambda^2}d\lambda. \tag{7.3}$$

Using the wave equation to substitute for ν we define

$$B_\lambda(T)d\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda k_BT} - 1}d\lambda. \tag{7.4}$$

Planck's function contains an important result, which is that wavelength corresponding to the peak value of B_λ is inversely proportional to temperature. Known as Wien's law, this provides a simple way to find the temperature of a blackbody.

$$\lambda_{max}T = 2.898 \times 10^6 \qquad [\text{nm K}]. \tag{7.5}$$

In reality we observe and measure the flux F, the power per unit area, which can be found by integrating B over solid angle. This integration shows that the flux F at frequency ν for a blackbody temperature T is given by $F_\nu(T) = \pi B_\nu(T)$.[4] Figure 7.2 shows the blackbody flux for different temperatures. As the temperature increases, the overall flux increases. and the peak of the distribution moves to higher frequencies.

In terms of wavelength this is written as $F_\lambda(T) = \pi B_\lambda(T)$. Figure 7.3 shows the blackbody emission as a function of wavelength for different temperatures. Here the observed solar flux is overplotted, and we find that it corresponds to a surface temperature of about 5780 K.

The Stefan-Boltzmann law describes how the total flux $F = \int F_\nu d\nu = \int F_\lambda d\lambda$ depends on the temperature, and was introduced in the previous lecture:

$$F = \sigma T^4, \tag{7.6}$$

where σ is Stefan-Boltzmann's constant. In reality, real stars are only approximate blackbodies, as we can see from Figure 7.3. The total emitted power is the luminosity, and so for a spherical star of radius R, its luminosity is $L = 4\pi R^2 F$. In this case, we can define an effective temperature T_{eff} which satisfies[5]

$$L_* = 4\pi R_*^2 \sigma T_{\text{eff}}^4. \tag{7.7}$$

[4] You may have been introduced to solid angles in Vector Fields and Electricity and Magnetism in year 1. Just like the circumference of a circle can be divided into 2π radians of angle, the surface of a sphere can be divided into 4π steradians of solid angle. Here the factor of 4 disappears, because there is a factor of $1/2$ for the fact that there is only outward radiation, and another factor of $1/2$ that accounts for the tilt between the surface normal and the line of sight (this is Lambert's cosine law).

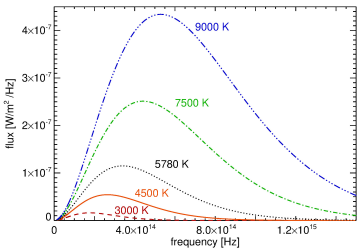

Figure 7.2: Flux $F_\nu(T) = \pi B_\nu(T)$ from a blackbody as a function of frequency for different temperatures.

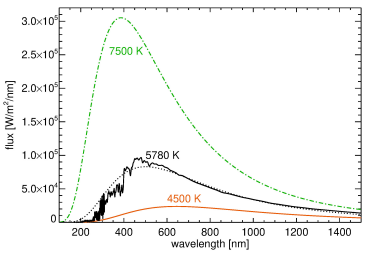

Figure 7.3: Similar to 7.2, but showing the flux $F_\lambda(T) = \pi B_\lambda(T)$ as a function of wavelength for different temperatures. Overplotted is the Sun's spectrum.

[5] This was introduced in the previous lecture.

7.3 Spectral Lines

7.3.1 Absorption and Emission Lines

The light from stars contains 'lines', where for specific frequencies the intensity is reduced when looking directly at the object. Essentially, material in the star's atmosphere absorbs light from the photosphere. This is then re-emitted in all directions, and so the intensity seen by the observer is reduced, as illustrated in Figure 7.4. In contrast, an observer looking at the star's atmosphere will see emission lines, where the atmosphere is 'glowing' at particular frequencies.

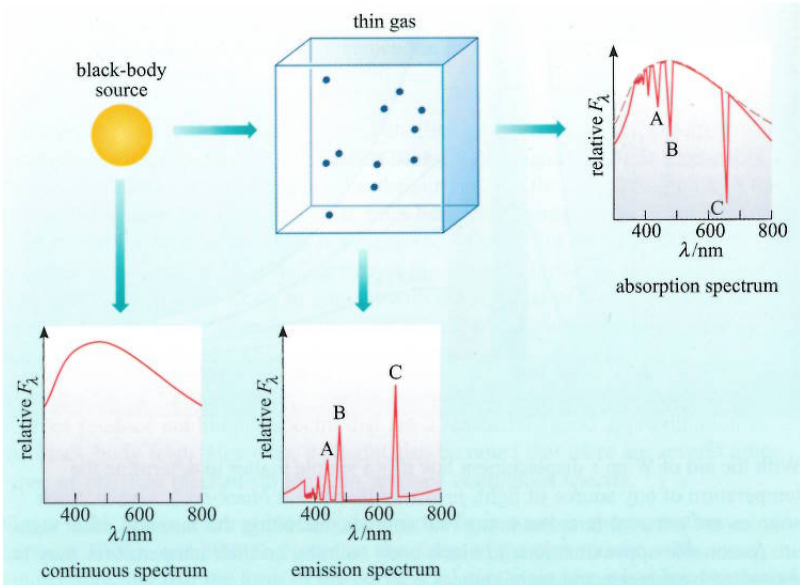

Figure 7.4: Diagram illustrating how absorption and emission lines arise, and the importance of the observer's look direction (taken from *An Introduction to the Sun and Stars* edited by S.F. Green and M.H. Jones).

7.3.2 Energy Levels and Transitions

The physics of spectral lines is associated with atomic physics. When a photon is absorbed, the photon's energy places an electron in a higher energy level; the wavelength of absorption therefore depends on the energy levels of different atoms. Here we illustrate this with hydrogen, the simplest system we can study. The energy levels of the hydrogen atom are given by

$$E_n = -\frac{m_e \left(e^2/4\pi\epsilon_0\right)^2}{2\hbar^2}\frac{1}{n^2} = -\frac{hc\,R}{n^2} = -\frac{13.6}{n^2} \quad [\text{eV}]. \qquad (7.8)$$

Here $\hbar = h/2\pi$, m_e is the electron mass, ϵ_0 is the permittivity of free space, e is the elementary charge, $R = 1.097 \times 10^7\ \text{m}^{-1}$ is the Rydberg constant. Converting the equation to electronvolts, 13.6 eV is the hydrogen ionisation potential. If the wavelength of the photon is ν, its energy is $h\nu$ where h is Planck's constant. If it causes the electron to move from energy level n to n', we can

write that $\Delta E = E_{n'} - E_n = h\nu$. This is illustrated in Figure 7.5. Excitations from $n = 1$ correspond to the Lyman series, which is in the ultraviolet. For example, Lyman-α is assocated with the $n = 1$ to $n = 2$ transition and occurs at 121.6 nm. Excitations from $n = 2$ correspond to the Balmer series which is in the optical. The so-called Hα line is at 656.3 nm, and corresponds to the $n = 2$ to $n = 3$ transition.

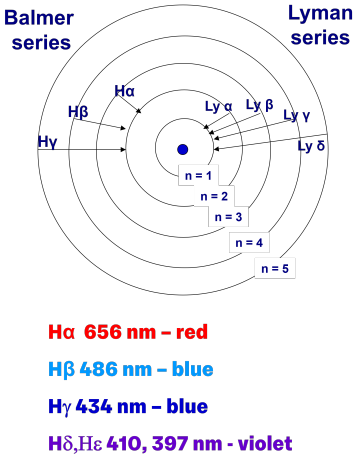

Figure 7.5: Cartoon to illustrate hydrogen's energy levels and associated emission. The Lyman series of lines results from photons de-excitation to the $n = 1$ state and the Balmer series from de-excitation to $n = 2$.

7.3.3 *Occurrence and Strength of Lines*

The occurrence and strength of lines depends on the relative abundance of different atoms, the temperature of the absorption medium, and the transition probabilities. This can be very complicated! Spectroscopy is a very powerful research tool allowing us to measure many physical properties including: photospheric chemical abundances; the radial velocity of a star from the Doppler shift of a spectral line; stellar rotation from additional line broadening; the measurement of mass inflow or outflow from asymmetries in line profiles; and the measurement of photospheric magnetic fields through the Zeeman effect.[6]

7.4 *The Structure of the Solar Atmosphere*

By carefully observing the Sun's atmosphere at different frequencies, we can deduce its structure, finding that it is divided into four layers: the photosphere, the chromosphere, the transition region, and the corona.[7] The structure of the solar atmosphere is shown in Figure 7.6, and images of the different layers are shown in the PowerPoint slides that accompany this lecture.

- Photosphere. Most of the light we see comes from the photosphere. It is only 500 km thick and its temperature decreases from ∼9000 K at the lower boundary to ∼4500 K at the top of the photosphere. The temperature minimum marks the transition to the chromosphere.

- Chromosphere. This region extends from the top of the photosphere to about 2000 km above it, but its extent and and temperature can vary. It can only be easily seen from Earth during a total eclipse when the Moon blocks out the light of the photosphere and the chromosphere appears as a narrow region with a reddish colour. It appears red because in the chromosphere, hydrogen atoms absorb energy radiated from the photosphere, then re-radiate as described above. In the visible, the predominant emission line is the Balmer Hα line at 656.3 nm.

- Transition region. Here the temperature rises very steeply from about 20,000 K to roughly 10^6 K in a distance of just $\sim$ 100 km. At these temperatures, hydrogen is essentially fully ionized so light emitted by the transition region is dominated by ions such

[6] Historical note: As summarised in e.g. Wikipedia, it also enabled the discovery of helium before it was isolated in the laboratory. Norman Lockyer, jointly credited for the discovery of helium, was the world's first professor of astronomical physics at the Royal College of Science which is of course part of Imperial College London. In other achievements, he also founded the journal Nature.

[7] Above the corona is the solar wind, which is beyond the scope of this module, but central to the year 4 space physics elective module.

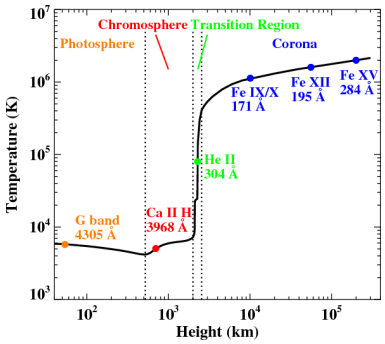

Figure 7.6: The temperature profile of the solar atmosphere, together with different relevant atomic lines. Note that e.g. Fe XV means 14 times ionised, as the Roman numeral I means a neutral atom. Reproduced from Yang et al., 2009, Figure 2.

as carbon, oxygen, and silicon, with emission in the ultraviolet that is only visible from space.

- Corona. This is the outermost, hottest, and most tenuous layer of the Sun's atmosphere, which extends out several solar radii.[8] The corona does not emit as a blackbody since it is not opaque, but we can estimate the wavelength of emission by equating the average photon energy $h\nu$ to the thermal energy $\sim k_B T$. Since $T \sim 10^6$ K, most of the corona's emission is at X-ray wavelengths. The lower layers of the corona produce X-ray and EUV emission lines from highly ionised minor ions like Iron.

[8] The NASA Parker Solar Probe is the first mission to 'touch' the Sun by flying through the corona.

The photosphere, chromosphere and corona exhibit a number of different features; these are summarised in table 7.1. Images illustrating each of these features are shown in the PowerPoint slides accompanying this lecture.

Region	Feature	Properties
Photosphere	Granules	Mark the upper parts of convection cells. The dark edges represent downflow. $\sim 10^3$ km across (lifetime $\sim$ 10 minutes).
	Supergranules	Their boundaries form where the convection causes magnetic field to be concentrated. $\sim 30,000$ km across (lifetime $\sim$ 12 - 24 h).
	Sunspots	Dark spots on the photosphere. Regions of concentrated magnetic field, which disrupt convection. In sunspots, the plasma is cooler, and therefore darker. Groups of sunspots form in pairs, with the magnetic field pointing out of one sunspot and into the other.
	Faculae	Bright regions formed by smaller-scale magnetic flux concentrations.
Chromosphere	Plage	Bright patches associated with concentrations of magnetic field; the counterpart of Faculae.
	Chromospheric Network	Web-like pattern associated with the Supergranule magnetic field structure.
	Prominences & Filaments	Dense clouds of material suspended by magnetic fields. Called prominences when seen on the limb of the Sun (e.g. during a total solar eclipse), otherwise known as filaments.
	Spicules	Small, jet-like eruptions in the chromospheric network, each lasting for only a few minutes.
Corona	Active Regions	Regions over Sunspots containing magnetic loop structures and significant time-varying activity. Often a location for solar flares (localised brightening across the electromagnetic spectrum)
	Coronal Loops	Closed magnetic field lines around sunspots and active regions; can last for days or weeks but can also suddenly become unstable and erupt (causing coronal mass ejections).
	Coronal Holes	Regions of 'open' magnetic field where the field line extends out into the Heliosphere.
	Helmet Streamers	Network of magnetic loops with dense plasma connecting the sunspots in active regions, typically occurring above prominences.
	Polar Plumes	Long thin streamers associated with open magnetic field lines at the poles.

Table 7.1: Important features seen in different regions of the solar atmosphere.

7.5 *The Magnetic Sun and the Solar Cycle*

The Sun's magnetic field permeates its atmosphere and shapes its appearance. Lower in the solar atmosphere the magnetic field can be measured by line splitting through the Zeeman effect. In the corona the line-broadening caused by the high temperature prevents the Zeeman effect being used, but the magnetic topology can be observed directly because of how it interacts with the plasma, forming prominences and loops and other coronal structure.

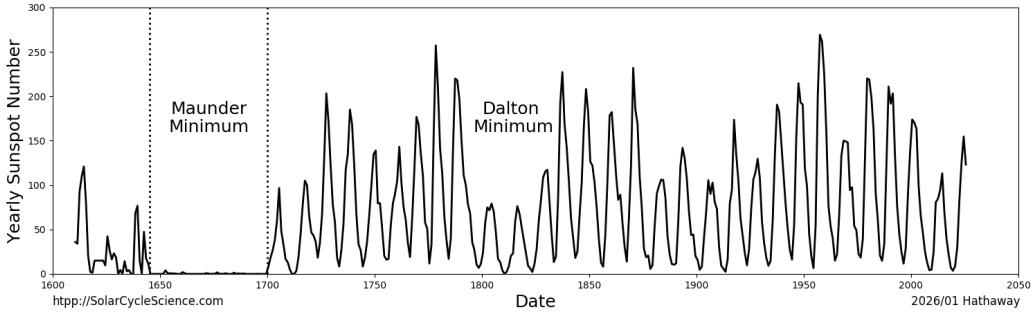

Figure 7.7: The number of sunspots observed on the Sun as observed since the early 1600's. Image taken from https://solarcyclescience.com/solarcycle.html.

The number of sunspots changes with time, with an 11 year period (see Figure 7.7). At solar minimum, very few, or indeed no sunspots are observed, whereas at solar maximum the number of sunspots peaks. There have also been periods of time with no visible cycle such as the Maunder minimum.

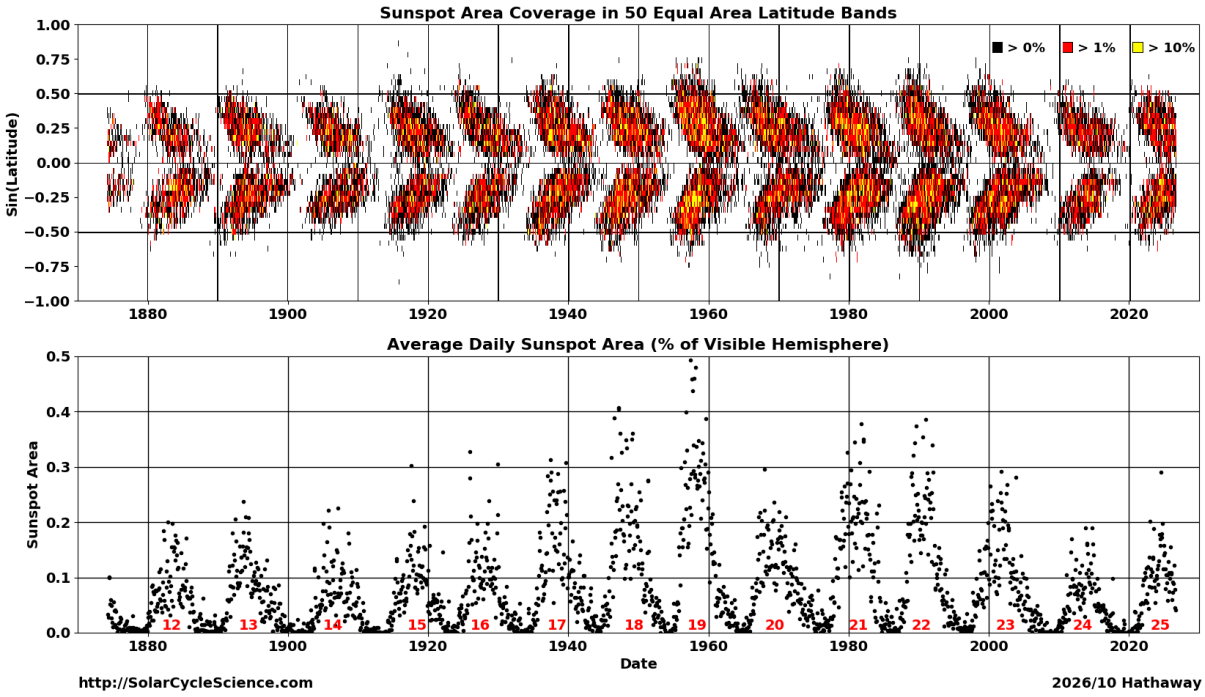

Figure 7.8: The top panel shows the latitudinal distribution of sunspots as a function of time since 1874 (the so-called Butterfly diagram). The bottom panel shows the total area of the visible Sun covered by sunspots. Image taken from https://solarcyclescience.com/solarcycle.html.

Some further key information about sunspots is shown in Figure 7.8. The top panel shows the latitudinal distribution of sunspots over time. This figure is often called the 'butterfly' diagram and reveals that sunspots emerge in two latitudinal bands which descend to the equator over several years. The bottom panel indicates how much of the photosphere is covered by sunspots as a function of time; this peaks every ∼11 years. Whilst progress has been made in recent years, the precise physics of the solar cycle is poorly understood and solving this remains an outstanding challenge in physics.

8
Observing Stars

8.1 Stellar Magnitudes

8.1.1 Apparent Magnitude

When we observe stars we measure their apparent brightness, which is classified by their 'apparent' magnitude. This follows a logarithmic scale, but note that brightness decreases as magnitude increases.[1] Apparent magnitude is represented by lower-case m. We define

$$m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right), \tag{8.1}$$

where m_1 and m_2 are the apparent magnitudes of object 1 and object 2, respectively. F_1 and F_2 are the fluxes of object 1 and object 2 measured at Earth, respectively. For example, if $m_1 = 5$ and $m_2 = 10$, then $F_1/F_2 = 10^{-5/-2.5} = 100$.

A reference object is needed, and the Vega magnitude system defines the apparent magnitude of the Vega star $m_{Vega} = 0$. This means

$$m_1 = -2.5 \log_{10} \left(\frac{F_1}{F_{\text{Vega}}} \right). \tag{8.2}$$

The apparent magnitudes of some different objects are shown in table 8.1.

[1] A lot of things in astronomy are done for historical reasons. This definition finds its origin in the work of the Greek astronomer Hipparchus, who divided stars based on visual brightness into 6 classes, with 1st magnitude being the brightest, and 6th magnitude being 100× dimmer.

Table 8.1: The magnitudes of various objects. Data taken from Las Cumbres Observatory.

Object	Magnitude
Sun	-26.7
Full Moon	-12.6
Venus at its brightest	-4.4
Sirius (α CMa)	-1.6
Vega (α Lyr)	0.0
Binocular Limit	+9.5
Pluto at its brightest	+13.7
Hubble Space Telescope limit	+30

8.1.2 The Parsec

However, rather than the apparent magnitude, we are more concerned with the intrinsic properties of the star. To access this, we need to establish an 'absolute' magnitude, which is defined at a standard fixed distance. This standard distance is related to the parsec, which is defined as illustrated in Figure 8.1. The parsec is the distance at which a length of 1 astronomical unit (1 AU) subtends an angle of 1 arcsec = $1'' = (1/3600)^\circ$. One parsec (1 pc) = $(3600 \times 180/\pi)$ AU = 206,265 AU = 3.09×10^{16} m = 3.26 light years.

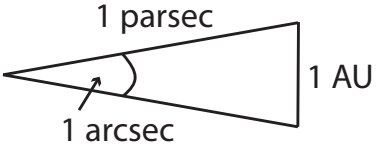

Figure 8.1: Definition of the parsec.

8.1.3 *Absolute Magnitude*

The absolute magnitude of a star is defined as its magnitude observed at distance of 10 pc. We write absolute magnitude using an upper-case M. Consider a star of Luminosity L, and known distance d_1 from Earth. Firstly we can write that

$$F_1 = \frac{L}{4\pi d_1^2}, \tag{8.3}$$

where F_1 is the star's flux at Earth. Its apparent magnitude is m where

$$m = -2.5 \log_{10}(F_1). \tag{8.4}$$

$\mathcal{F}_1$ is the flux received were the star at distance of D = 10 parsecs,

$$\mathcal{F}_1 = \frac{L}{4\pi D^2}, \tag{8.5}$$

and by definition

$$M = -2.5 \log_{10}(\mathcal{F}_1). \tag{8.6}$$

Using equations 8.3 and 8.5, we can write that

$$F_1 d_1^2 = \mathcal{F}_1 D^2. \tag{8.7}$$

Substituting for $\mathcal{F}_1$ in equation 8.6

$$M = -2.5 \log_{10} \left(\frac{F_1 d_1^2}{D^2} \right), \tag{8.8}$$

$$M = -2.5 \log_{10}(F_1) - 5 \log_{10} \left(\frac{d_1}{D} \right). \tag{8.9}$$

Using equation 8.4,

$$\boxed{M = m - 5 \log_{10} \left(\frac{d_1}{D} \right)} \tag{8.10}$$

where D = 10 pc. If d_1 is given in units of parsec, this simplifies to

$$M = m - 5 \log_{10}(d_1) + 5. \tag{8.11}$$

8.1.4 *Observational Passbands*

Real telescopes and observation systems consist of a detector, together with some level of filtering sensitive to a specific wavelength band. Although several systems exist, the Johnson-Cousins system is commonly used to define wavelength bands and is summarised in Table 8.2, as well as shown in Figure 8.2. This means that we can define magnitudes in different wavelength bands, for example m_V would be the apparent visual magnitude, and M_I would be the absolute infrared magnitude of an object. For reference, the Sun's absolute visual magnitude is $M_V = 4.83$.

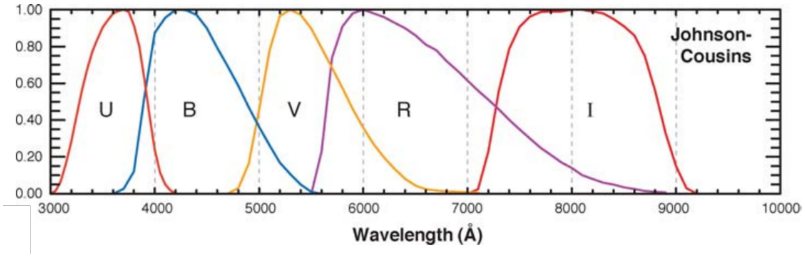

Figure 8.2: Schematic showing the passbands of five colour bands in the Johnson-Cousins system, adapted from Bessell, 2005, Figure 1.

Filter	Name	Central wavelength (nm)	Width (nm)
U	Ultraviolet	366.3	65
B	Blue	436.1	89
V	Visual	544.8	84
R	Red	640.7	158
I	Infrared	798.0	154

Table 8.2: The Johnson-Cousins UBVRI filter system. Data taken from Bessell, 2005, Table 1

8.1.5 *Bolometric Magnitude*

The bolometric magnitude of an object ($m_{\rm bol}$: apparent, $M_{\rm bol}$: absolute) corresponds to its flux summed over all wavelengths. We write

$$m_{\rm bol} = m_{\rm V} + BC \tag{8.12}$$

where BC is the bolometric correction. Different values of BC apply for corrections to m_I, m_U, etc. The bolometric correction is generally negative since $m_{\rm bol}$ is smaller than the magnitude in any given passband. Note also that the bolometric correction is intrinsic to the star, and so can be applied to both absolute and apparent magnitudes.[2] The Sun's absolute bolometric magnitude = 4.74.

[2] You should convince yourself that $BC = m_{\rm bol} - m_{\rm V} = M_{\rm bol} - M_{\rm V}$ using equation 8.10.

8.2 *Stellar Colours and Temperatures*

Stars have different colours because they have different temperatures. This follows directly from Planck's function and Wien's law, where cooler stars are more red, and hotter stars are more blue. This means that different stars have different magnitudes in different passbands, as illustrated in Figure 8.3.

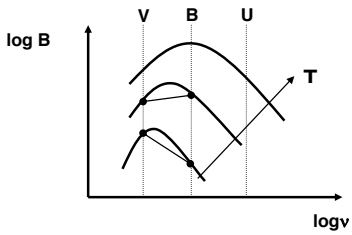

Figure 8.3: Three curves show Planck's function for increasing blackbody temperature. The vertical lines mark the centre of the V, B, and U passbands. The coolest star will have $m_V < m_B$ (it is brighter in the Visual passband), whereas the other two stars have $m_V > m_B$ (they are brighter in the Blue passband).

By measuring the magnitude in two passbands (e.g. V and B), we may calculate the difference, known as the 'colour index'. For example, the $B - V$ index = $m_B - m_V$ (or $M_B - M_V$). For the coolest star in Figure 8.3, there is more emission the Visual passband, and so m_V will be *smaller* than m_B. This means that $B - V$ will be positive for this star. In contrast, for the next hottest star in Figure 8.3, $B - V$ will be negative, because m_B is less than m_V. Colour indices can be used to estimate the effective temperature, T_{eff}, of the star. This estimate is distance independent, and efficient, but less accurate than other methods we will discuss later in the module.

Stars are conventionally classified according to their colour, or spectral type, summarised in Table 10.1. In addition to the letter

classification, there is a numerical subclass, ranging from 0 to 9, with 0 being the hottest. This means that for example an F9 star and a G0 star are quite similar. The Sun is a G2 star. From the spectral type we can determine a star's average photospheric temperature to an uncertainty of about 5%.

Type	Colour	T_{eff} (K)	Main characteristics	Examples
O	Blue	>25000	He^+ lines; strong UV	Mintaka
B	Blue-white	11000-25000	Neutral He lines	Rigel, Spica
A	White	7500-11000	Strong H lines	Sirius, Vega
F	Yellow-white	6000-7500	Weak metal lines	Procyon
G	Yellow	5000-6000	Sun-like spectrum	Sun, Capella
K	Orange	3500-5000	Metal lines dominate	Arcturus, Aldebaran
M	Red	<3500	Molecular bands noticeable	Betelgeuse, Antares

Table 8.3: The spectral classification of stars into stellar types.

8.3 Distance: Trigonometric Parallax

As the Earth orbits the Sun, the position of nearby stars relative to very distant stars appears to change. This is because of the parallax effect illustrated in Figure 8.4. If the parallax angle, p, is measured in arcseconds and the distance, d, is measured in parsecs then

$$\boxed{d = \frac{1}{p}} \quad \text{for } d \text{ in parsec and } p \text{ in arcsec.} \tag{8.13}$$

For example, the parallax angle of Alpha Centauri is $p = 0.76''$, so $d \sim 1.32$ pc.

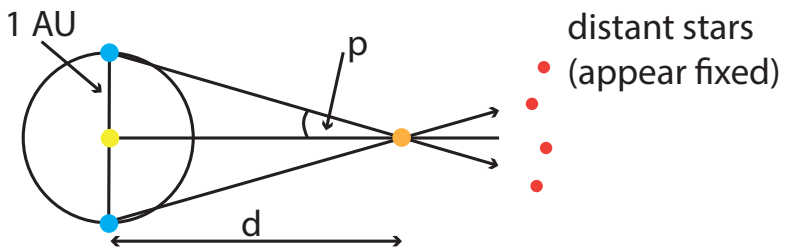

Figure 8.4: Illustration of parallax. The Earth (blue) orbits the Sun (yellow), and the relative position of the target star (orange) changes with respect to the fixed distant stars (red).

8.4 Proper Motion

The proper motion of a star refers to its change of angular position over time, independent of parallax effects. This is illustrated in figure 8.5. We can measure this angle, μ, over some period of time Δt, and if we know the distance to the star, d, (for example by using parallax), then the transverse speed is

$$v_t = \frac{d \sin(\mu)}{\Delta t}. \tag{8.14}$$

For small angles $\sin(\mu) = \mu$. We observe that proper motion tends to decrease as d increases, so the proper motion of a star can

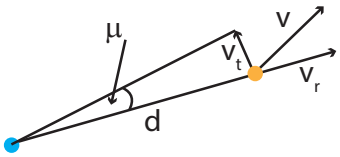

Figure 8.5: Illustration of proper motion.

be used as a rough indicator of the star's distance, if the distance is otherwise unknown.

The radial speed of a star, v_r, can be found by looking for Doppler shifts in spectral lines. If an expected wavelength λ is observed at λ', then

$$v_r = c\left(\frac{\lambda' - \lambda}{\lambda}\right). \tag{8.15}$$

If the line is redshifted to longer wavelengths, it is moving away from the observer. In contrast, a blueshift indicates the object is moving towards us. Finally, the overall speed of the object is v where

$$v = \sqrt{v_t^2 + v_r^2}. \tag{8.16}$$

Star motions are not simply random, and the existence of a super-massive black hole at the centre of our galaxy was revealed by the orbits of stars around Sagittarius A*.[3]

[3] Reinhard Genzel and Andrea Ghez were awarded the Nobel Prize in Physics in 2020 for this work.

9
Stellar Binary Systems

9.1 Binary Star Orbits

Two thirds of stars are observed in binary orbits, and this allows us to measure their mass. The two stars in a binary system occupy elliptical orbits around their common centre of mass, as shown in Figure 9.1. Star S_1 has mass M_1 and occupies an orbit with semi-major axis a_1. Star S_2 has mass M_2 and its orbit has semi-major axis a_2. It can be shown that $M_1 a_1 = M_2 a_2$, and so if $M_1 > M_2$, then $a_1 < a_2$. Thus the more massive star occupies a smaller orbit, as might be expected.

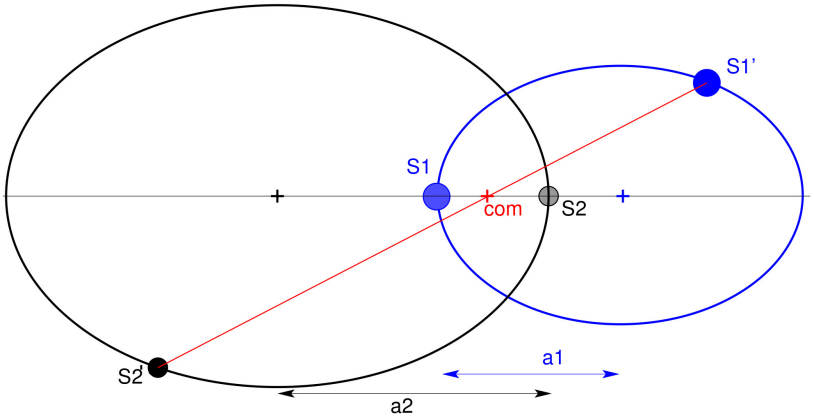

Figure 9.1: Diagram showing the elliptical orbits of two stars in a binary system. Note that the line connecting the two stars always passes through the system centre of mass.

To make the analysis more straightforward, we will consider circular orbits for simplicity (see Figure 9.2). Here the stars S_1 and S_2 again have mass M_1 and M_2, respectively, and orbital radii of r_1 and r_2 respectively. The separation between the two stars is $r = r_1 + r_2$ which is constant. Both stars have the same orbital period P and angular frequency ω where

$$\omega = \frac{2\pi}{P}. \tag{9.1}$$

For star S_1,

$$\frac{GM_1M_2}{r^2} = M_1 r_1 \omega^2, \tag{9.2}$$

and for star S_2,

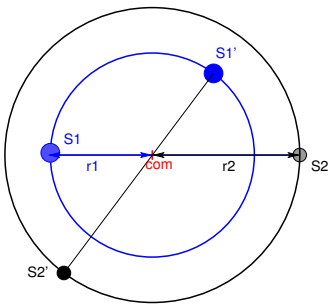

Figure 9.2: Cartoon of a binary orbit where both objects occupy circular orbits.

$$\frac{GM_1M_2}{r^2} = M_2 r_2 \omega^2. \tag{9.3}$$

The left-hand side of equations 9.2 and 9.2 are identical, so therefore $M_1 r_1 = M_2 r_2$ (Thus defining the centre of mass). Adding equation 9.2 (divided by M_1) and equation 9.3 (divided by M_2) we find that

$$\frac{G}{r^2}(M_2 + M_1) = \omega^2(r_1 + r_2) = \omega^2 r = \frac{4\pi^2}{P^2} r. \tag{9.4}$$

If we then write $M = M_1 + M_2$ where M is the total mass of the system we find

$$M = \frac{4\pi^2}{G}\frac{r^3}{P^2} \tag{9.5}$$

Equation 9.5 also holds for elliptical orbits, but instead of r, it uses $a = a_1 + a_2$. If M_1 or M_2 increase then the period P decreases, whereas if r (or a) increases, then P increases. This is Kepler's third law in the context of a binary system.

9.2 *Inclined Observing Geometries*

In reality, as illustrated in Figure 9.3, the binary system will be inclined to the observer. This is particularly important for spectroscopic observations (see below). We characterise this with the angle i. In this case, the maximum line-of-sight ('radial') velocities, v_1 and v_2 of stars S_1 and S_2 are then given by

$$v_1 = \frac{2\pi r_1 \sin i}{P} \quad \text{and} \quad v_2 = \frac{2\pi r_2 \sin i}{P}. \tag{9.6}$$

Figure 9.3: Diagram showing the observing geometry when the plane of the binary system is inclined to the view direction.

Note here that $r_1/r_2 = v_1/v_2$.

We now start with equation 9.5, which we show here again:

$$M = \frac{4\pi^2}{G}\frac{r^3}{P^2}. \tag{9.7}$$

We first write

$$r = r_1 + r_2 = r_1\left(1 + \frac{M_1}{M_2}\right) = r_1\frac{M}{M_2}. \tag{9.8}$$

This allows us to substitute for r in equation 9.7:

$$M = \frac{4\pi^2}{GP^2}\frac{M^3}{M_2^3} r_1^3. \tag{9.9}$$

We then multiply by $\sin^3 i$:

$$M \sin^3 i = \frac{4\pi^2}{GP^2}\frac{M^3}{M_2^3}(r_1 \sin i)^3. \tag{9.10}$$

We now have the equation in a format where we can use equation 9.6 to write

$$M \sin^3 i = \frac{M^3}{M_2^3}\frac{4\pi^2}{GP^2}\left(\frac{v_1 P}{2\pi}\right)^3. \tag{9.11}$$

Simplifying this expression,

$$M \sin^3 i = \frac{M^3}{M_2^3} \frac{P}{2\pi G} v_1^3, \tag{9.12}$$

and so

$$\boxed{\frac{(M_2 \sin i)^3}{M^2} = \frac{P}{2\pi G} v_1^3.} \tag{9.13}$$

Equation 9.13 is the binary mass function. An equivalent equation connects M_1 and v_2:

$$\frac{(M_1 \sin i)^3}{M^2} = \frac{P}{2\pi G} v_2^3. \tag{9.14}$$

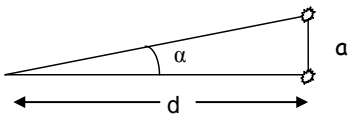

Figure 9.4: The physical and angular separation of an idealised binary system where both stars occupy a circular orbit.

[1] A third star, Proxima Cen, was thought to be a third member of the system at a much greater distance but recent observations suggest it may not be gravitationally bound to the binary system.

9.3 *Measuring Stellar Properties*

There are several different approaches to identifying the properties of stellar binary systems, and in reality a combination of these methods would be used as feasible and appropriate.

9.3.1 *Visual Binaries*

In a visual binary system both stars can be seen as separate points of light. By measuring a_1/a_2, the mass ratio M_1/M_2 can be found directly. The semi-major axis, a, can be found from the angular separation α and the distance d, where

$$a = d \sin(\alpha) \simeq d\alpha. \tag{9.15}$$

and illustrated in Figure 9.4. By measuring the period P, we find M via equation 9.5. Once M and M_1/M_2 are known, the individual masses can be established. α Centauri is an example of a visual binary. α Cen A and α Cen B are separated by $\sim$ 23 AU, at a distance of 1.338 pc, and a system period of $\sim$ 80 years.[1]

Finally, in an astrometric binary system, only one star is visible. The best known example of an astrometric binary is Sirius, where a wobble in the proper motion of Sirius A was ascribed to the existence of an unseen companion Sirius B which is now known to be a dim white dwarf.

9.3.2 *Spectroscopic Binaries*

In a spectroscopic binary system, the two stars are too close together to be resolved as separate points of light. In this case, we observe periodic shifts in spectral lines, which are caused by the Doppler effect and associated with the orbital motion of each star. In this way, P is found relatively easily from observations, as illustrated in Figure 9.5. This method also reveals the radial velocity.

In a double-lined binary system, the spectroscopic signatures of both stars can be observed, and P, v_1 and v_2 can all be measured.

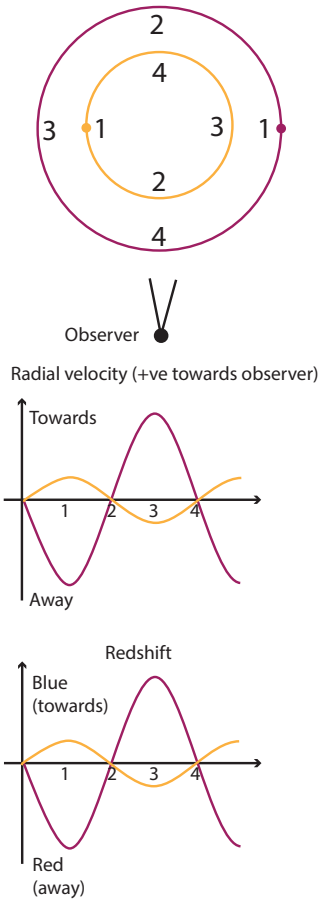

Figure 9.5: Detection of a binary system through spectroscopy. At point 1, the radial velocity of star S_1 towards the observer is maximised. At point 2 the radial velocity of S_1 is 0, becoming negative at point 3, and 0 at point 4. Star S_2 shows the opposite behaviour. The blue- and redshift follows the radial velocity.

If the inclination angle i is known, then the masses M_1 and M_2 can also be found. If i is unknown, then the minimum mass is found by setting $\sin i = 1$.

In a single-lined binary, one of the stars (e.g. S_2) is too faint to see its signature in spectroscopic observations. As a result it is only possible to measure P and v_1, and the mass function is derived from their values.

9.3.3 *Eclipsing Binaries*

If the inclination angle is close to 90°, then it is possible that from the point of view of the observer, the stars regularly pass in front of each other and successively eclipse each other, as illustrated in Figure 9.6. During each eclipse there is a reduction in the total flux received from the system.

When the brighter, primary, star (yellow in Figure 9.6) is eclipsed by the fainter, secondary, star (orange in Figure 9.6), this is known as the primary eclipse. When the fainter star is eclipsed by the brighter star, the light is reduced but not as much and this is known as the secondary eclipse. We might expect the brighter star will be larger, because luminosity scales with mass.

More specifically, if the primary star S_P with visible area A_P produces flux F_P, and the secondary star S_S with visible area A_S produces flux F_S, then the total flux of the system is $F_T = F_P + F_S$. If $A_P > A_S$, then during a primary eclipse, the flux from the primary star is reduced and the total flux is given by the following formula:

$$F_{T,primary} = F_S + F_P - \frac{A_S}{A_P} F_P = F_S + F_P \left(1 - \frac{R_S^2}{R_P^2}\right). \tag{9.16}$$

During a secondary eclipse, the flux is simply $F_{T,secondary} = F_P$, as the secondary star is completely obscured. The eclipse durations gives estimate of stellar radii, and it is often also possible to derive the temperatures by changes in the luminosity at different pass-bands. Finally, if the velocities of the stars can be measured, then their masses can be calculated as well.

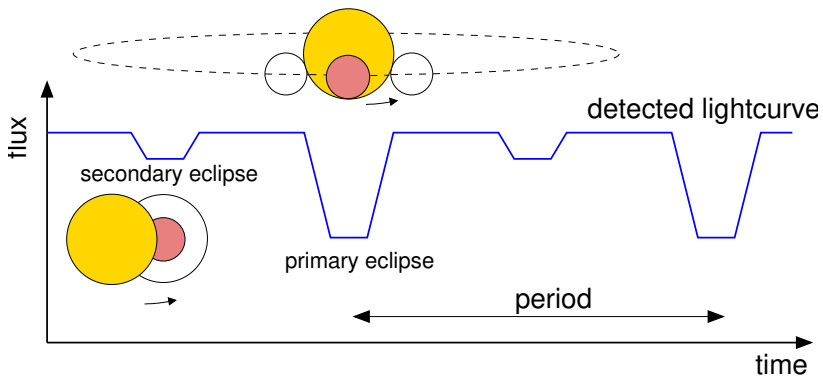

Figure 9.6: Diagram showing the change in flux from the binary system as a consequence of each star being successively eclipsed by the other.

9.4 *Interacting Binary Systems*

In a binary system, the Roche lobe of each star demarcates where material is gravitationally bound to that star. The radius of a star tends to increase with age[2] and if one star expands sufficiently, its outermost material may extend outside its Roche lobe and be more attracted to its companion star. In this case, mass and angular momentum is transferred from one star to the other. These systems are known as interacting binaries, and this interaction may affect the evolution of both stars.

Interacting binary systems can host a wide variety of interesting astrophysics. For example, milli-second pulsars are formed when the transfer of angular momentum from a larger companion star spins up a neutron star to the extent that its rotation period is less than $\sim$ 10 msec, as illustrated in Figure 9.7.

A second important phenomenon is the type Ia supernova. This occurs when mass accretes onto a white dwarf. If the mass of the white dwarf reaches a critical limit, then it becomes unstable and explodes. This critical limit helps to make the type Ia supernova a so-called 'standard candle' as its peak luminosity is relatively well understood and predictable.[3]

[2] As we will explore in the next two lectures.

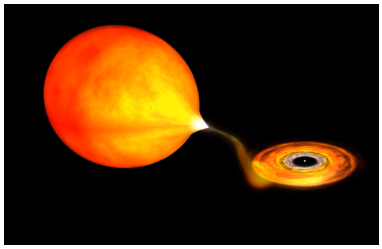

Figure 9.7: Still image from a visualization of a binary system containing a millisecond pulsar. (Credit NASA Science Visualization Studio)

[3] Observations of type Ia supernovae were critical to the discovery that the expansion of the universe is accelerating; this was awarded the Nobel prize in physics 2011. They are also a major target for the Nancy Grace Roman Space Telescope.

10
Classifying Stars and the Hertzsprung-Russell Diagram

10.1 How to Classify Stars

In general we classify stars according to their temperature (i.e. their colour)[1], and their luminosity (i.e. their brightness). We will also discover that this may connect to other properties like mass and radius in an ordered way, reflecting the behaviour we observed in the scaling laws in lecture 6.

The Russell-Vogt theorem states that the equilibrium structure of an ordinary star is determined uniquely by its mass and chemical composition, and argues that given a particular set of initial conditions, this sets the properties of the star. While not universally true, it is reasonable to assume this for the majority of stars we observe.

[1] The method of classifying stellar types based on their colour, or spectra, was devised by Annie Jump Cannon between 1918 and 1924.

10.2 Spectral Classification: Colour and Temperature (Revisited)

In lecture 8, section 8.2, we started to explore the relationship between colour and temperature, based on the fact that stars emit as blackbodies. Spectroscopy provides another very powerful tool to probe the photospheric temperature of stars, because different star types have different spectral lines associated with their atmospheres.[2]

At the most basic level, we might expect that stars with lower surface temperature (and therefore appearing redder) have cooler atmospheres, and therefore it is easier for lines to be observed from neutral atoms and even molecules, as they are more likely to be present in significant concentrations. In hotter stars we would expect much more ionisation, and therefore hydrogen and helium spectral lines to be present. Information about spectral lines associated with different stellar types is summarised in Table 8.3. Figure 10.1 shows graphically how different lines are more or less prominent at different temperatures, and therefore in different stellar types.

In summary, we have identified three methods that can be used

[2] This was first identified by Cecelia Payne-Gaposchkin; the Institute of Physics medal in plasma, solar and space physics is named for her.

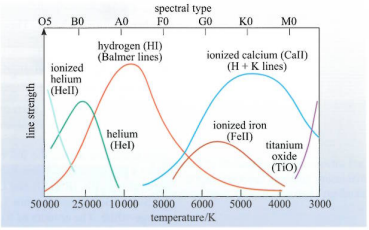

Figure 10.1: This diagram shows how different spectral lines have different strengths for different temperatures, which correspond to the different stellar spectral types. For example, ionised calcium lines are more prominent in K-type stars, whereas hydrogen Balmer lines are more prominent in A-type stars. (Figure reproduced from An Introduction to the Sun and Stars, edited by S. F. Green and M. H. Jones.)

to estimate the photospheric temperature of a star.

- Wien's law uses λ_{max} of the blackbody spectrum fit to estimate the effective temperature. This is rather crude, but simple.
- The photometric method uses the difference in magnitude between different filter bands. It is good for surveys, and assessing large numbers of stars, but it is approximate.
- Spectral classification is most accurate, but requires each star's spectrum to be measured and studied in detail.

10.3 Luminosity Classification

The second property of a star that it is critical to characterise is the luminosity. Two stars may have the same temperature and colour, but could be vastly different in terms of their luminosity, and therefore their underlying physics. More specifically, recall that

$$L_* = 4\pi R_*^2 \sigma T_{\text{eff}}^4, \tag{10.1}$$

where L_* is the star luminosity, R_* is the radius and T_{eff} is the stellar effective temperature. This means that for two stars with the same temperature, the luminosity is proportional to R_*^2.

In lecture 1 we found that the surface gravity $g \propto M/R^2$. This means for a large star, g is lower, which in turn means that the atmosphere is not as tightly bound. Large stars therefore have lower atmospheric pressure. In contrast, g is higher for small stars, so their atmospheres are more dense with higher pressures.

This physical difference provides a way to distinguish between the two stars, using a phenomenon called collisional or pressure broadening of spectral lines. If the pressure of the star's atmosphere is high, this causes the width of the spectral lines to increase. At higher pressure there are more collisions and interactions between the photons and the material in the stellar atmosphere, which distorts the energy levels of the interacting atoms, allowing a wider range of photon wavelengths for each transition.[3]

[3] Note this is not the same as Doppler broadening caused by thermal motions.

Consequently, in a large star with lower atmospheric pressure, the effect of pressure broadening is low, and the spectral lines are narrow. In contrast, a small star with a higher atmospheric pressure will exhibit significant pressure broadening in its spectral lines.

In conclusion, for a given temperature a small star that is less luminous will exhibit more pressure broadening of its spectral lines, whereas a large star that is more luminous will exhibit less pressure broadening. Evidently, the correlation is not precise, and in fact luminosity is classed as follows[4]:

[4] Here we use Roman numerals.

- Ia and Ib: these are the most luminous stars, known as supergiants, evolved stars with very inflated radii ($\sim 100\ R_\odot$).
- II: Luminous giant stars.
- III: Giant stars.

- IV: Sub-giants.
- V: Dwarf stars. Note that all main sequence stars regardless of their mass are luminosity class V, even a 20 $M_\odot$ star.

This classification system groups stars by luminosity for a given effective temperature. It is important to realise that a main sequence star at a high temperature could be significantly more luminous than a giant star that is bigger but happens to be at a lower temperature.

This pressure broadening relationship has been calibrated using stars at known distances, where the luminosity can be found from the absolute magnitude. Once this is known, then it can be applied to stars at unknown distances. In fact, armed with the measurements of a star's temperature (from any of the three methods in the previous section), luminosity (estimated from pressure broadening), and an estimate of its radius (calibrated based on nearby similar stars whose distance is known), the distance to the star can be estimated.

10.4 Types of Stars

The full designation of a star's classification includes its spectral type and its luminosity class. For example the Sun is classified as a G2 V star and Betelgeuse is an M2 Ia star, an M2 supergiant.

The European Space Agency Gaia mission has played a pivotal role in mapping the properties of stars. Capable of making measurements to an accuracy of a few microarcseconds, Gaia allowed the distances to stars across the entire Milky Way Galaxy to be measured for the first time. For example, the 2022 data release reported colour and parallax measurements for 1.5 billion stars as well as precise magnitude, position and proper motion determinations.

Generally speaking, there are a few major classes of stars. These include:

- The main sequence: 90% of the stellar population, powered by hydrogen fusion.
- Red dwarfs: the smallest main sequence stars, extremely long-lived.[5]
- Giant stars: Large, luminous stars, above the main sequence. These may also be referred to as red giants.
- Supergiant stars: Even larger and more luminous stars, also above the main sequence.
- White dwarfs: small less luminous stars, below the main sequence.
- Neutron stars, pulsars and magnetars: super-dense objects. Pulsars are rapidly rotating neutron stars that emit pulses of electromagnetic radiation, typically radio but also other wavelengths.

[5] If the mass of the object is too low, it will form what is known as a brown dwarf. Here, the mass is below $< 0.08\ M_\odot$, at which point the rate of internal energy generation by fusion is insufficient to match the rate of energy radiation from the surface.They emit predominantly in the infrared. They are more massive than the most massive planets like Jupiter, and, unlike planets, seem to form from the interstellar medium in the same way as stars.

For the main sequence population, masses range from 0.1 - $\sim$ 100 $M_\odot$, although the maximum is not firmly known.

10.5 Hertzsprung-Russell (HR) Diagrams

Hertzsprung-Russell (HR) diagrams plot the luminosity of stars as a function of their temperature, and reveal both the physics at work inside stars and how stars evolve over their lifetime. There is not one single HR diagram. An observational HR diagram might plot, for example the absolute visual magnitude[6] as a function of colour index (e.g. $B - V$). An example of such a diagram is shown in Figure 10.2.

[6] It can be appropriate to use apparent magnitude, if all the stars under consideration are approximately the same (unknown) distance away. This would be relevant for studying a star cluster, for example.

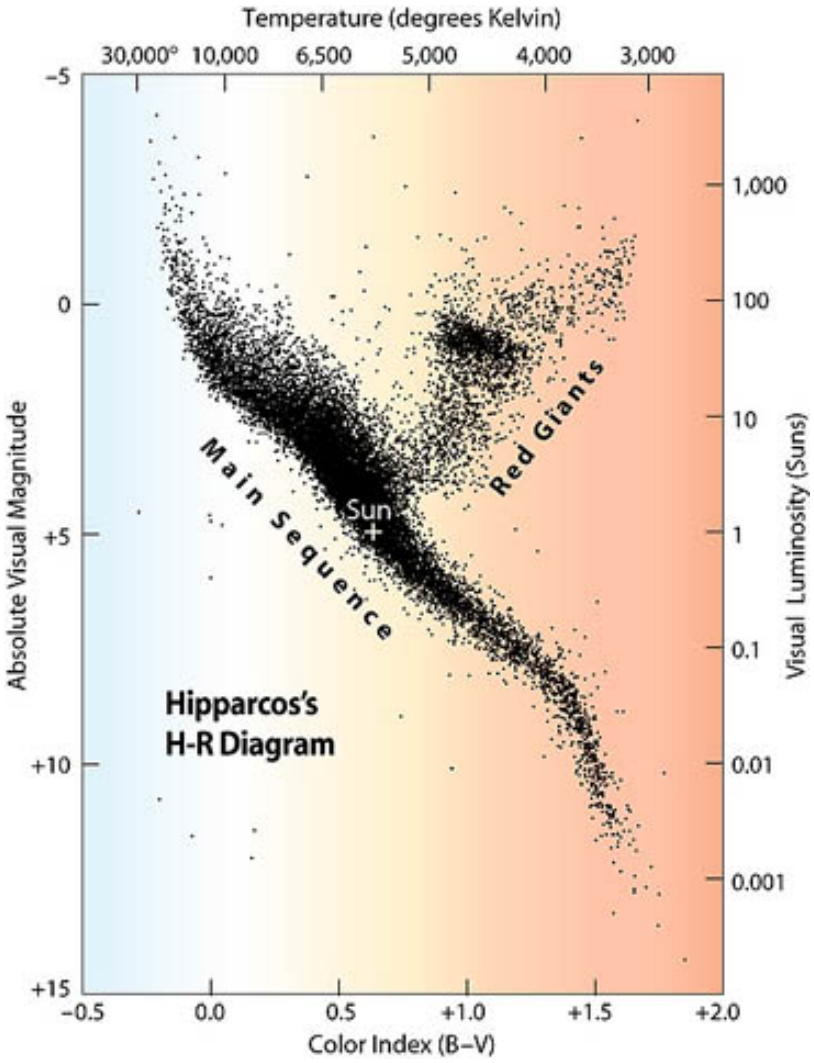

Figure 10.2: An HR diagram obtained from the ESA Hipparcos mission, where absolute visual magnitude is plotted as a function of $B - V$ colour index. Note that bluer colours are to the left - so **temperature decreases when moving to the right!** The magnitude decreases from bottom to top, because smaller magnitude objects are more luminous. (Image credit: ESA)

Another version of the HR diagram (perhaps appropriate from a more theoretical approach), is to plot luminosity against effective temperature. At each point the radius of the star is uniquely defined as $R = \sqrt{L_*/4\pi\sigma T_{\rm eff}^4}$. A star that is very luminous but very cool has to be very large, and is thus given the name red

giant. A sketch of a luminosity-temperature diagram is shown in Figure 10.3; the diagonal red lines indicate lines of constant radius. On a log-log scale, the gradient of these lines is -4, since $d\log_{10} L/d\log_{10} T_{\rm eff} = 4$.

In summary we observe that the main sequence occupies a line running from top left to bottom right on an HR diagram. Above the main sequence are the giants and supergiants. Below the main sequence are smaller less luminous stars called white dwarfs.

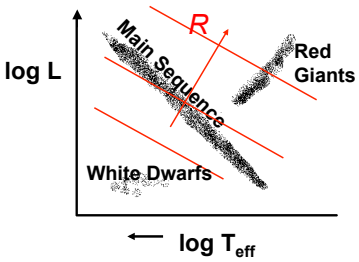

Figure 10.3: Another version of an HR diagram, here plotting luminosity as a function of effective temperature. Note that a log-log scale is used, and **temperature decreases from left to right**. Lines of constant radius are marked in red.

10.6 Scaling of Luminosity and Temperature

Recall that in Lecture 6 we found that for lower mass stars where the pp-chain operates, the Luminosity $L_* \propto T_{\rm eff}^{5.6}$. In problem sheet 2 you will have found that in fact it can scale approximately as $L_* \propto T_{\rm eff}^{4}$. However, in higher mass stars where the CNO cycle operates, $L_* \propto T_{\rm eff}^{8}$. We expect therefore to see a break in the slope of the main sequence on an HR diagram, where it is steeper for more massive stars.

11
The Life (and Death) of Stars

11.1 The Distribution of Stars on the HR Diagram

The locations and numbers of stars on an HR diagram tell us a great deal about the lifecycle of stars. From our knowledge of the physics, we infer that stars have a finite life. The fact that we can see stars being born[1] means that not all stars were created at the same time, so the stars we see today are not all at the same stage of their evolution. This togther with the fact that stars are grouped in particular places on the HR diagram suggests that they have distinct stages in their evolution. An implication of this idea is that a star moves around the HR diagram as it evolves.

[1] e.g. in the Orion nebula

We expect to observe more stars in evolutionary phases that last a long time. Only a few stars should be observed in phases that last a short time. Therefore, regions of the HR diagram with lots of stars correspond to long-lived evolutionary phases. On this basis, we conclude that the main sequence is where a star spends most of its life, since 90% of all stars are found there. Other regions with significant populations are the giant, supergiant and white dwarf regions, also indicating that stars spend a not-insignificant time in these forms.

However, we must be careful in our interpretation. The absence, or low number, of stars in a particular region of the HR diagram could be because physics doesn't allow stars to have that combination of luminosity and temperature, or because there are certain viewing restrictions (e.g. dust), or because stars do not spend a large fraction of their lifespan in that state.

One way to explore this is by considering the population of nearest objects, and the population of brightest objects in the sky. In the near-object population, where we survey just a small part of the galaxy, we mainly observe common stars (largely the main sequence), and less luminous stars like white dwarfs are also apparent. For example, Figure 11.1 shows the stars observed by Gaia up to 100 pc distance. In the bright-object population, we observe very luminous main-sequence stars, but also very luminous (class I, II) stars that are very far away.

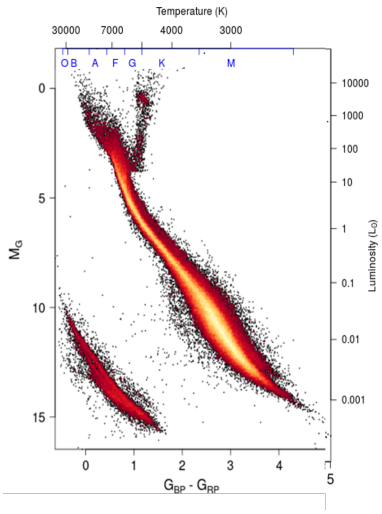

Figure 11.1: HR diagram obtained from Gaia data (Gaia collaboration, Babusiaux et al. 2018, A&A 616, A10). Most of the stars are on the main sequence, but the red-giant branch and the white-dwarf branch here can also be seen. Note that this is only out to 100 pc, and so rarer, distant, very bright stars will not be included in this dataset.

11.2 Mass and the HR Diagram

Figure 11.2 shows the typical masses of stars for different populations on the HR diagram. Overall, we observe stars with masses from 0.08 $M_\odot$ to ∼100 $M_\odot$. Higher-mass stars are rarer, whereas low-mass stars are more common.[2] We observe the following

- Main sequence: mass increases with L and $T_{\rm eff}$, as expected from scaling relations. homology solutions to stellar structure. The mass range is from 0.1 $M_\odot$ - 50 $M_\odot$, i.e. a 500× increase in mass. This gives roughly a 10^{10} increase in luminosity since $L \propto M^{3.5}$.
- Red giants: masses are typically a few solar radii, comparable to stars on the lower end of the main sequence. There is no correlation with luminosity or photospheric $T_{\rm eff}$.
- Supergiants: have greater masses than red giants, being of the order of 8 - 20 $M_\odot$. This is comparable with masses of stars on the upper main sequence. Again there is no ordering with luminosity or temperature.
- White dwarfs: all have masses less than about 1.5 $M_\odot$, and as with supergiants and red giants mass is not ordered with luminosity or temperature.

[2] However, low-mass stars are not very luminous so harder to observe and potentially under-count.

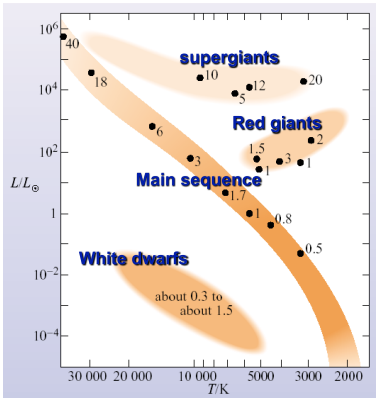

Figure 11.2: Typical masses of stars at various positions on the HR diagram. (*An Introduction to the Sun & Stars* eds. Green & Jones).

There is thought to be an upper limit to the mass of a star. As a star becomes more massive, its internal temperature rises (following the scaling we found in previous lectures). Recalling that the radiation pressure $P_{rad} = \frac{1}{3}aT^4$, whereas the gas pressure $P_{gas} = nkT$ we find:

$$\frac{P_{\rm rad}}{P_{\rm gas}} = \frac{aT^3}{3nk}. \tag{11.1}$$

Equation 11.1 shows that as temperature increases, radiation pressure contributes much more strongly. It can be shown[3] that this leads to an estimate of the Eddington luminosity which sets an upper mass limit for a star.[4]

To make connections between different regions on the HR diagram, we must consider if stars change mass significantly throughout their lifetime. Whilst main sequence stars, giants and supergiants all exhibit stellar winds, over their lifetime the mass loss is insignificant. On the other hand, we also observe structures called planetary nebula, as shown in Figure 11.3.[5]. In fact this is the outer shell of the star that has been ejected, and this is a significant mass loss process.

[3] See Problem Sheet 3

[4] The exact upper limit of stellar mass is still unknown; there is good evidence for stars with masses up to about 150 $M_\odot$.

[5] The name is historical, given by William Herschel, but planets are irrelevant here.

Figure 11.3: The Cat's eye Nebula (NGC6453), a Planetary Nebula. The central star is an O-type star (a Wolf-Rayet star), and is expected over the next million years to become a white dwarf (Credit: Hubble Space Telescope).

11.3 The Life (and Death) of Stars

Starting at the beginning of a star's life, the **Zero Age Main Sequence** (ZAMS) curve is defined by values of L and $T_{\rm eff}$ corresponding to static, homogeneous stars that have just started burn-

ing hydrogen. ZAMS stars are well described by our equations of stellar structure.

In the lower main sequence, (M <1.5 $M_\odot$), hydrogen burning in the core is through the pp chain. Energy transport from the core is radiative, with a convective envelope further out. In the upper main sequence (M >1.5 $M_\odot$) the CNO cycle takes over because the core temperatures are higher and in these conditions it releases energy at a much higher rate than the p-p chain. This causes convective instability in the central regions of higher mass stars, surrounded by a radiative envelope, as illustrated in Figure 11.4.

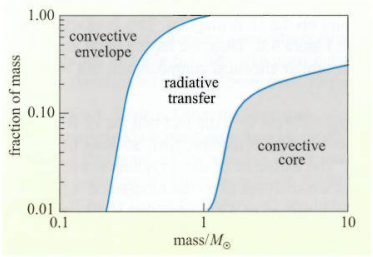

Figure 11.4: The structure of a star depends on its initial mass. Low-mass stars are entirely convective, at intermediate masses there is a radiative zone with an outer convective zone, and in high mass stars the core is convective with an outer radiative envelope. From *An Introduction to the Sun & Stars* eds. Green & Jones.

A star does not change much in luminosity or effective temperature during its main sequence phase. If it did, we would not find the ordered range of masses that we observe. However, a star's luminosity does increase slightly as it fuses hydrogen, and so it will move slightly upwards on the HR diagram, meaning that the main sequence is a band, not a narrow line.

After the main sequence phase, less massive stars (less than $\sim 8M_\odot$) become red giants. When hydrogen fuel in the star's core starts to be exhausted, the core pressure reduces and the star collapses. The increased pressure allows helium to start fusing into carbon further powering the star. The net result is that the star expands into a red giant.

Eventually the red giant becomes unstable, ejecting its outer layers in a planetary nebula, and leaving behind a white dwarf. This is the core of the original star which steadily cools as it is no longer generating energy. Electron degeneracy pressure acts against gravity.

In contrast, for more massive stars on the main sequence (between $\sim 8M_\odot$ and $\sim 20M_\odot$), fusion in the core can sequentially involve heavier elements (H, then He, C, Ne, O, Si and Fe). Fusion beyond iron does not release energy, and the loss of internal pressure leads to violent collapse and rebound, causing a type II supernova, leaving behind a neutron star.[6]

As remarked upon in Lecture 10, a neutron star is a super-dense object composed primarily of neutrons. Pulsars are rapidly rotating neutron stars that emit pulses of electromagnetic radiation, typically radio but also other wavelengths.[7] A magnetar is a highly magnetised neutron star.

[6] If the core is more massive than ∼3 solar masses, it is thought that this will create a black hole...

[7] This provided the first evidence for neutron stars.

11.4 Stellar Mass and Lifetime

As derived in Lecture 6, the lifetime of a main sequence star scales with its mass, where we found

$$\tau_{MS} \propto \frac{energy}{L_*} \propto \frac{M}{L_*} \propto M^{-2.5}. \tag{11.2}$$

This means that the amount of time a star spends on the main sequence before it turns into a red giant or supergiant depends on its mass. We can test this scaling by looking at HR diagrams of star

clusters. We assume that all the stars in the cluster are formed from the same material and formed at the same time. This means that by looking at the range of masses seen in main sequence stars in the cluster, we can infer the cluster's age.

There are two main types of cluster. Open clusters (e.g. the Pleiades) are not strongly gravitationally bound, typically found within our galaxy, and consist of $10^2 - 10^5$ stars. In contrast, globular clusters (e.g. M13) are more tightly bound, can be seen both in our galaxy and nearby galaxies, and consist of 10^6 stars. Comparing the HR diagrams of the Pleiades and M13 (shown in Figure 11.5 and Figure 11.6, respectively), we can see that the Pleiades is younger, because it has higher luminosity (and therefore higher mass) stars on the main sequence. In contrast, M13 contains many red giants and in fact its age is estimated to be ∼14 Gyr, having formed quite soon after the Big Bang.

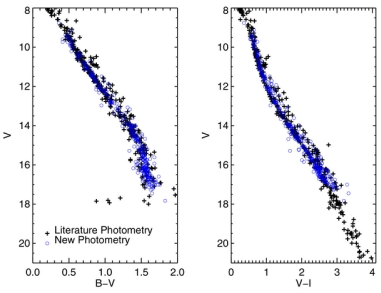

Figure 11.5: HR diagrams of the Pleiades cluster (Kamai et al, 2014, Astrophys. J., 148, 30).

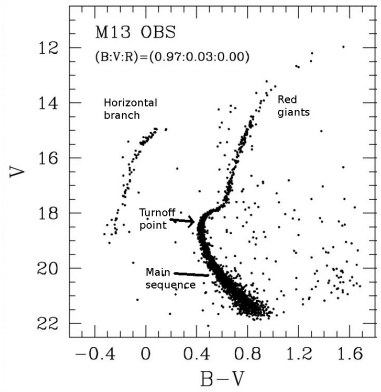

Figure 11.6: HR diagram of the M13 globular cluster (Rey et al 2001, Astrophys. J., 122, 3219). Note that the distribution of stars on the main sequence is very different to Figure 11.5, with no main sequence stars below visual magnitudes of about 18.

11.5 *Cluster Age and the Main Sequence Turn Off*

As a cluster ages, its stars exit the main sequence in an orderly fashion, starting with the most massive and ending with the least massive. This means that the line of main sequence stars on the HR diagram gets shorter over time, with the upper-left end of the line moving down and to the right. This point is known as the 'main sequence turn off'. The location of the main sequence turn-off can be used to estimate the age of the cluster.

If the turn off point occurs when hydrogen is exhausted, we may write that

$$\tau_t = E_{\mathrm{MS}}/L \tag{11.3}$$

where τ_t is the lifetime, E_{MS} is the energy used and L is the luminosity. The energy used is

$$E_{\mathrm{MS}} = E_{Hkg} \times f \times X \times M. \tag{11.4}$$

In this equation, $E_{Hkg} \sim 6.58 \times 10^{14}\,\mathrm{J/kg}$ is the energy/unit mass released by fusion, f is the fraction of hydrogen burnt, X is the fraction of hydrogen in the star, and M is the mass of the star. We define $\xi = E_{Hkg} \times f \times X$ and so

$$E_{\mathrm{MS}} = \xi M \tag{11.5}$$

As a result, we can estimate the Cluster age as τ_t where

$$\tau_t = \frac{E_{\mathrm{MS}}}{L} = \xi \frac{M}{L} \tag{11.6}$$

Since $L \propto M^{3.5}$, this means that $M \propto L^{2/7}$ and $M = M_\odot \left(L/L_\odot\right)^{2/7}$. Using equation 11.5 to substitute for M,

$$\tau_t = \xi \frac{M_\odot}{L} \left(\frac{L}{L_\odot}\right)^{2/7} \tag{11.7}$$

Further manipulating this equation we find[8]

$$\tau_t = \xi \frac{M_\odot}{L_\odot} \left(\frac{L}{L_\odot} \right)^{-5/7} \tag{11.8}$$

It is convenient to normalise age to the expected age of the Sun $\tau_{t\odot}$ where

$$\tau_{t\odot} = \xi \frac{M_\odot}{L_\odot} \left(\frac{L_\odot}{L_\odot} \right)^{-5/7} \tag{11.9}$$

and $\tau_{t\odot} \sim 10$ bn years. Using equation 11.8 and equation 11.9, and assuming ξ is the same for all stars (e.g. fraction of hydrogen burnt $f = 0.2$ and fraction of hydrogen in the star $X = 0.6$), we find

$$\frac{\tau_t}{\tau_{t\odot}} = \left(\frac{L}{L_\odot} \right)^{-5/7}. \tag{11.10}$$

In summary, the age of the cluster τ_t is defined by the luminosity of the brightest star on the main sequence in the cluster. This star sits at the main sequence turn off. From this we can measure $\log L/L_\odot$ and use equation 11.10 to find the cluster age.

[8] You should write this out by hand to convince yourself if in doubt.